Three problems in kernel methods: proofs and finite-size measurements

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Abstract

Three questions about kernel methods are answered here, and a fourth is measured. For the Gaussian kernel with a fixed bandwidth on the sphere, the minimum-norm interpolant of noisy labels is benign along the sequence $d_\ell = 2^{2^\ell}$, $n_\ell = 2^{2^{2\ell}}$: its excess risk tends to zero in probability, and almost surely along the sequence, for every fixed-degree spherical-polynomial target (Theorem 5.1). At the exponential sample size $n = 2^d$ the same interpolant is obstructed: the expected excess risk on pure noise is at least $\frac{5}{36}\sigma^2$ in the limit, for every fixed bandwidth (Theorem 4.1). For ridge regression on two augmented views of each sample through a random feature map with an odd activation, the optimised objective and the clean prediction risk have the same limits in law as those of a correlated Gaussian surrogate that keeps the shared latent variable of the two views (Theorem 6.1); the proof rests on an exact Gaussianisation of each block conditional on the feature matrix. The hard edge of the degree-two Veronese matrix remains open; the two lemmas that stand between its reduction and the hard edge are measured, and the measurements point to a sharper statement, that the leave-one-out normal has a semicircular spectrum. The proofs are accompanied by finite-size measurements of the quantities they turn on: the leverage certificate of the lower bound, the harmonic dimension gap that separates the benign regime from the obstructed one, and the leverage concentration behind the equivalence. Two further results are evaluated on data: an exact same-sample risk identity for a kernel correction fitted on top of a learned mean, on the EMIT and OCO-2 radiative-transfer emulation corpora, where its sign depends on the component being emulated and the correction fitted to out-of-fold residuals gains nothing; and the compressed-resolvent hypothesis behind growing-rank spiked singular vectors, whose compression error along rays of fixed rank-to-size ratio approaches $\sqrt{r/n}$ plus a term of order r/n.

1 Introduction

Kernel ridge regression and its ridgeless limit are among the few learning methods whose high-dimensional theory is close to complete, and several natural questions about them have stayed open. This paper settles three of them and measures a fourth. The questions are stated precisely in Section 2; they concern the minimum-norm interpolant of the Gaussian kernel on the sphere at two very different sample sizes, and ridge regression on augmented random features.

Along the doubly exponential sequence $d_\ell = 2^{2^\ell}$, $n_\ell = 2^{2^{2\ell}}$ the interpolant is benign for every fixed-degree spherical-polynomial target and every fixed bandwidth (Theorem 5.1, Section 5); the mechanism is a dimension gap, polynomial in d on both sides of the critical harmonic degree.

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At the exponential sample size $n = 2^d$ the interpolant of pure noise is obstructed: its expected excess risk is at least $\frac{5}{36}\sigma^2$ in the limit (Theorem 4.1, Section 4); the proof is an expectation argument built on the leverage of a block of harmonic degrees, with no concentration step. For two augmented views of each sample passed through a random feature map with an odd activation, the optimised ridge objective and the clean prediction risk have the same limits in law as those of a correlated Gaussian surrogate (Theorem 6.1, Section 6); conditional on the feature matrix there is a Gaussian block whose second moments match the actual block exactly, and the replacement argument then only has to pay for the fluctuation of a 2×2 leverage matrix. The fourth question, the hard edge of the degree-two Veronese matrix, is open; Section 9 states the reduction that isolates it and measures the two lemmas that the reduction names.

Part II measures, at sizes a computer can reach, the objects the proofs are made of. A proved constant may be far from the truth: between $d = 9$ and $d = 13$ the quantity bounded in Theorem 4.1 measures between 1.70 and 1.88 times σ^2 at $h = 1$ and between 0.54 and 0.64 at $h = 1/2$ (Section 7), so the theorem is not tight, and the measurement says by roughly how much. A lemma in a reduction can be checked before it is proved: both lemmas that the reduction of Target 4 names look true at finite size, one of them in a stronger form than the reduction needs, and the other with its constant identified as the fourth moment of the semicircle law (Section 9). An exact identity can be confronted with data directly: Section 11 does this for a same-sample risk identity on two radiative-transfer emulation corpora, where it separates the components on which a residual kernel correction helps from the one on which it hurts. Finite d cannot settle a $\liminf_{d\to\infty}$, and the sample sizes of Theorem 5.1 leave the reachable world at $\ell = 3$; where a table rests on a single seed or a single corpus, the text says so.

Part I
Theorems and proofs

2 The problems

Throughout, $X_1, \dots, X_n$ are i.i.d. uniform on S^{d-1}, K is the Gaussian kernel

$$K(x, y) = \exp\Big(- \frac{\|x - y\|^2}{2h^2} \Big) = e^{-\gamma} e^{\gamma\langle x,y\rangle}, \qquad \gamma = h^{-2},$$

with $h > 0$ fixed, $N_{d,s}$ is the dimension of the degree-s spherical harmonics, P_s is the zonal harmonic normalized so $P_s(1) = 1$ [3], and $\lambda_{d,s}$ are the Mercer eigenvalues [40], so that $K(x, y) = \sum_s \lambda_{d,s} N_{d,s} P_s(\langle x, y\rangle)$ and $\tau_s = N_{d,s}\lambda_{d,s}$ satisfies $\sum_s \tau_s = 1$.

Target 1 (correlated Gaussian equivalence). Two augmented views of each sample, $x_{ij} = \sigma(F^\top(g_i + \Delta z_{ij}))$, $j = 1, 2$, with a linear teacher $t_i = \xi^\top g_i$ and the ridge objective $L(w) = \frac{1}{4n}\sum_i \sum_j (t_i - x_{ij}^\top w)^2 + \frac{\lambda}{2}\|w\|^2$. Does the optimized objective, and the clean test risk of its minimizer, agree with those of a matched Gaussian surrogate? This is a correlated form of the Gaussian equivalence of random-feature models [23, 28, 39].

Target 2 (benign overfitting). Along $d_\ell = 2^{2^\ell}$, $n_\ell = 2^{2^{2\ell}}$, with fixed h, Gaussian noise and any fixed-degree spherical-polynomial target of bounded L^2 norm, does the conditional excess risk of the minimum-norm interpolant tend to zero in probability? Benign interpolation with kernels is a high-dimensional phenomenon [35, 46, 6, 22, 37]; the question is whether it survives a bandwidth that does not scale with the dimension.

Target 3 (the obstruction). With $n = 2^d$ and zero signal, is $\liminf_{d\to\infty} \mathbb{E}\,\mathcal{E}_{2^d,d} \geq c(h)\sigma^2$ for an explicit $c(h) > 0$?

Target 4 (the Veronese hard edge). With $p = d(d+1)/2$ and $z(x) = \big((x_a^2 - 1)/\sqrt{2}\big)_a \oplus (x_a x_b)_{a<b}$, $x \sim N(0, I_d)$, does $p\, s_{\min}(Z)^2$ for the square $p \times p$ matrix Z with rows $z(X_i)$ obey the real square Wishart hard edge, whose limit law is $1 - e^{-\sqrt{t}-t/2}$ [19, 53]?

Theorems 5.1, 4.1 and 6.1 below answer Targets 2, 3 and 1. The proof of Target 2 also covers every target in the empirical low block below the critical degree, and this enlargement is sharp as a uniform bounded-L^2 class statement. Target 4 is open; Section 9 gives its reduction. Two further results established in [45] are evaluated on data in Part II: an exact identity for the change in population risk when a residual kernel correction is fitted on the same sample as the mean it corrects (Section 11), and the compressed-resolvent hypothesis behind the theory of growing-rank spiked singular values and vectors [8, 4] and its consequences for pruning (Section 12).

3 Harmonic preliminaries

Throughout, $X, X_1, \dots, X_n$ are independent and uniform on S^{d-1}, $d \geq 3$, and

$$K(x,y) = \exp\Big(-\frac{\|x-y\|^2}{2h^2}\Big) = e^{-\gamma} e^{\gamma\langle x,y\rangle}, \qquad \gamma = h^{-2},$$

with $h > 0$ fixed. We write $\mathcal{H}_{d,s}$ for the space of spherical harmonics of degree s on S^{d-1}, $N_s = N_{d,s} = \dim \mathcal{H}_{d,s}$, and $\Pi_s(x,y) = \sum_{a=1}^{N_s} \psi_{s,a}(x)\psi_{s,a}(y)$ for its reproducing kernel with respect to the uniform probability measure, so that $\Pi_s(x,y) = N_s P_s(\langle x,y\rangle)$ with P_s the zonal harmonic normalised by $P_s(1) = 1$ [3]. The addition theorem gives $\Pi_s(x,x) = N_s$, and orthogonality of distinct degrees gives, for every $x, z \in S^{d-1}$,

$$\mathbb{E}_Y\big[\Pi_s(x,Y)\Pi_t(Y,z)\big] = \delta_{st}\,\Pi_s(x,z), \qquad \mathbb{E}_Y \Pi_s(x,Y) = 0 \quad (s \geq 1). \tag{1}$$

The Mercer expansion of the kernel is $K(x,y) = \sum_{s\geq 0} \lambda_s \Pi_s(x,y)$ with $\lambda_s = \lambda_{d,s} > 0$ [40]; since $K(x,x) = 1$, the trace masses $\tau_s = N_s\lambda_s$ satisfy $\sum_s \tau_s = 1$. For a sample we write $K = (K(X_i,X_j))_{ij}$ for the Gram matrix, $\Pi_s(X,X) = (\Pi_s(X_i,X_j))_{ij}$, and $K_s = \lambda_s \Pi_s(X,X)$, so that $K = \sum_s K_s$. Each $\Pi_s(X,X) = \Phi_s\Phi_s^\top$ with $\Phi_s = (\psi_{s,a}(X_i))_{i,a} \in \mathbb{R}^{n\times N_s}$, and $\|\Phi_s\|_F^2 = \sum_i \Pi_s(X_i,X_i) = nN_s$.

Lemma 3.1 (harmonic multiplicities)**.** *For $d \geq 3$ and $s \geq 0$,*

$$N_{d,s} = \binom{d+s-1}{s} - \binom{d+s-3}{s-2} = \frac{(2s+d-2)\,(s+d-3)!}{s!\,(d-2)!},$$

$$\rho_s := \frac{N_{d,s+1}}{N_{d,s}} = \frac{(d+2s)(d+s-2)}{(s+1)(d+2s-2)}.$$

The ratio ρ_s is at least one and is non-increasing in s. For $s \leq d$ one has $\frac{3}{4}\binom{d+s-1}{s} \leq N_{d,s} \leq \binom{d+s-1}{s}$, and for $s^2 = o(d)$, $N_{d,s} = \frac{d^s}{s!}\, e^{O(s^2/d)}$.

Proof. The closed forms are classical [3]. From the product form, $\rho_s = \frac{d+s-2}{s+1} \cdot \frac{d+2s}{d+2s-2}$; both factors are non-increasing in s, and $\frac{d+s-2}{s+1} \geq 1$ for $d \geq 3$. The ratio of the two binomial coefficients is $\frac{s(s-1)}{(d+s-1)(d+s-2)} \leq \big(\frac{s}{d+s}\big)^2 \leq \frac{1}{4}$ for $s \leq d$, which gives the two-sided bound. For $s^2 = o(d)$, $\binom{d+s-1}{s} = \frac{d^s}{s!}\prod_{j=1}^{s-1}(1+j/d) = \frac{d^s}{s!}e^{O(s^2/d)}$ and the subtracted term is smaller by the factor $O(s^2/d^2)$. □

Lemma 3.2 (Mercer eigenvalues of the Gaussian kernel)**.** *With* $(d)_m := d(d+2)\cdots(d+2m-2)$ *and* $(d)_0 = 1$,

$$\lambda_{d,s} = e^{-\gamma} \sum_{r \geq 0} \frac{\gamma^{s+2r}}{2^r\, r!\, (d)_{s+r}} = e^{-\gamma}\, \Gamma\!\left(\tfrac{d}{2}\right) \left(\frac{2}{\gamma}\right)^{d/2-1} I_{s+d/2-1}(\gamma), \tag{2}$$

where I_ν *is the modified Bessel function of the first kind. Consequently, for all* $s \geq 0$,

$$\frac{\lambda_{d,s+1}}{\lambda_{d,s}} \leq \frac{\gamma}{d+2s}, \qquad \lambda_{d,s} \geq \frac{e^{-\gamma}\gamma^s}{(d)_s}, \qquad \tau_s \leq e^{-\gamma} \sum_{m \geq s} \frac{\gamma^m}{m!}. \tag{3}$$

In particular the eigenvalues are strictly decreasing in s *as soon as* $d > \gamma$.

Proof. Write $K(x,y) = e^{-\gamma} \sum_m \frac{\gamma^m}{m!} \langle x, y\rangle^m$. The Funk–Hecke formula [3] gives the degree-s coefficient of $t \mapsto t^m$ on S^{d-1}; equivalently, for $G \sim N(0, I_d)$ and the homogeneous harmonic extension $H_s(ru) = r^s Y_s(u)$ of $Y_s \in \mathcal{H}_{d,s}$, the identity $\mathbb{E}[H_s(G) e^{t\langle x, G\rangle - t^2/2}] = t^s H_s(x)$ and the comparison of the coefficient of t^{s+2r} give $\mathbb{E}[\langle x, G\rangle^{s+2r} H_s(G)] = \frac{(s+2r)!}{2^r r!} H_s(x)$, while $\mathbb{E}\|G\|^{2(s+r)} = (d)_{s+r}$. Dividing out the radial moments yields the series in (2); the Bessel form is the series $I_\nu(\gamma) = \sum_r \frac{(\gamma/2)^{\nu+2r}}{r!\,\Gamma(\nu+r+1)}$ [18, 10.25.2] rewritten with $\nu = s + d/2 - 1$. The first two bounds in (3) are termwise: the r-th term of λ_{s+1} equals $\frac{\gamma}{d+2(s+r)}$ times the r-th term of λ_s, and $\frac{\gamma}{d+2(s+r)} \leq \frac{\gamma}{d+2s}$; the second is the $r = 0$ term. For the third, expand $\langle x, y\rangle^m = \sum_{s \leq m} c_{m,s} P_s(\langle x, y\rangle)$ with $c_{m,s} \geq 0$ (the Gegenbauer coefficients of a monomial are non-negative) and $\sum_s c_{m,s} = 1$ by evaluating at $\langle x, y\rangle = 1$; hence $\tau_s = e^{-\gamma} \sum_{m \geq s} \frac{\gamma^m}{m!} c_{m,s} \leq e^{-\gamma} \sum_{m \geq s} \frac{\gamma^m}{m!}$. □

Lemma 3.3 (hypercontractivity at growing degree)**.** *For every* $Y_s \in \mathcal{H}_{d,s}$, *every* $s \geq 0$ *and every* $p \geq 2$, $\|Y_s\|_{L^p(S^{d-1})} \leq (p-1)^{s/2} \|Y_s\|_{L^2(S^{d-1})}$. *In particular, with* $r_s := \Pi_s / N_s$ *and* $z \in S^{d-1}$ *fixed,* $\|r_s(X, z)\|_{L^{2q}} \leq (2q-1)^{s/2} N_s^{-1/2}$ *for every* $q \geq 1$.

Proof. Let $G \sim N(0, I_d)$ and $H_s(G) = \|G\|^s Y_s(G/\|G\|)$, a homogeneous harmonic polynomial of degree s and hence an element of the s-th Gaussian chaos. Nelson's hypercontractive inequality [42, 29] gives $\|H_s(G)\|_p \leq (p-1)^{s/2} \|H_s(G)\|_2$. The radius and the direction of G are independent, so $\|H_s(G)\|_p = \|\,\|G\|^s\|_p \, \|Y_s\|_{L^p}$ and $\|H_s(G)\|_2 = \|\,\|G\|^s\|_2 \, \|Y_s\|_{L^2}$, and $\|\,\|G\|^s\|_p \geq \|\,\|G\|^s\|_2$ for $p \geq 2$. Dividing gives the first claim; the second follows from $\|r_s(\cdot, z)\|_{L^2}^2 = \Pi_s(z,z)/N_s^2 = 1/N_s$ by (1). □

4 Target 3: the exponential-sample lower bound

Here $n = 2^d$, the signal is zero, $Y_i = \varepsilon_i$ with $\varepsilon \sim N(0, \sigma^2 I_n)$, and $\hat{f}(x) = k(x)^\top K^{-1} \varepsilon$ is the minimum-RKHS-norm interpolant, $k(x) = (K(x, X_i))_i$. The Gaussian kernel is strictly positive definite, so K is invertible almost surely. The excess risk is $\mathcal{E}_{n,d} = \mathbb{E}_{X_0}[\hat{f}(X_0)^2]$, and we write $V(X) = \mathbb{E}_\varepsilon[\mathcal{E}_{n,d} \mid X]$.

Theorem 4.1 (Target 3)**.** *For every fixed* $h > 0$,

$$\liminf_{d \to \infty} \frac{\mathbb{E}\, \mathcal{E}_{2^d, d}}{\sigma^2} \;\geq\; \frac{5}{36},$$

the expectation being over the design and the noise.

The proof has three steps: an exact reduction of $V(X)$ to the leverage of a block of harmonic degrees, a lower bound on the expected leverage of a block, and the location of the last block whose dimension does not exceed n.

Lemma 4.2 (reduction to a block)**.** *Let* $G = \mathbb{E}_{X_0}[k(X_0)k(X_0)^\top] = \sum_s \lambda_s K_s$, $t_s = \operatorname{Tr}(K^{-1}K_s)$ *and* $q_s = \operatorname{Tr}(K^{-1}K_sK^{-1})$. *Then* $V(X)/\sigma^2 = \operatorname{Tr}(K^{-1}GK^{-1}) = \sum_s \lambda_s q_s$, $\sum_s t_s = n$, $t_s^2 \le nN_s\lambda_s q_s$ *for every* s, *and for every finite set* B *of degrees, with* $T_B = \sum_{s\in B} t_s$ *and* $D_B = \sum_{s\in B} N_s$,

$$\frac{V(X)}{\sigma^2} \;\ge\; \frac{T_B^2}{nD_B}. \tag{4}$$

Proof. $\mathbb{E}_\varepsilon \hat f(x)^2 = \sigma^2 k(x)^\top K^{-2}k(x)$, so $V(X)/\sigma^2 = \operatorname{Tr}(K^{-1}GK^{-1})$, and by (1),

$$G_{ij} = \sum_{s,t} \lambda_s\lambda_t\, \mathbb{E}_{X_0}\big[\Pi_s(X_i,X_0)\Pi_t(X_0,X_j)\big] = \sum_s \lambda_s^2 \Pi_s(X_i,X_j),$$

i.e. $G = \sum_s \lambda_s K_s$; hence $V/\sigma^2 = \sum_s \lambda_s q_s$, and $\sum_s t_s = \operatorname{Tr}(K^{-1}K) = n$. Since $K_s = \lambda_s \Phi_s\Phi_s^\top$, we have $t_s = \lambda_s\langle K^{-1}\Phi_s, \Phi_s\rangle_F$ and $q_s = \lambda_s\|K^{-1}\Phi_s\|_F^2$, so by Cauchy–Schwarz

$$t_s \le \lambda_s \|K^{-1}\Phi_s\|_F\, \|\Phi_s\|_F = \sqrt{\lambda_s q_s}\,\sqrt{nN_s}.$$

Summing over B and applying Cauchy–Schwarz once more, $T_B \le \sqrt{nD_B}\big(\sum_{s\in B}\lambda_s q_s\big)^{1/2} \le \sqrt{nD_B\, V(X)/\sigma^2}$. □

Lemma 4.3 (expected leverage of a block)**.** *Let* $d > \gamma$ *and* $L \ge 0$, *and put* $D = \sum_{s\le L} N_s$, $\lambda = \lambda_{d,L}$, $\tau_{>L} = \sum_{s>L}\tau_s$ *and* $T_{\le L} = \sum_{s\le L} t_s$. *Then*

$$\mathbb{E}\, T_{\le L} \;\ge\; \frac{nD}{D + n - 1 + \tau_{>L}/\lambda}. \tag{5}$$

Proof. Put $A = \sum_{s\le L} K_s$, $C = K - A = \sum_{s>L} K_s \succeq 0$, and $A_w = \lambda\sum_{s\le L}\Pi_s(X,X)$. By Lemma 3.2 the eigenvalues decrease in s, so $\lambda_s \ge \lambda$ for $s \le L$ and $0 \preceq A_w \preceq A$. For $\epsilon > 0$ let $K_\epsilon = K + \epsilon I$ and $C_\epsilon = A_w + C + \epsilon I$; then $K_\epsilon \succeq C_\epsilon \succ 0$, so $K_\epsilon^{-1} \preceq C_\epsilon^{-1}$ by operator monotonicity of the inverse. Since $A + C + \epsilon I = K_\epsilon$,

$$\operatorname{Tr}(K_\epsilon^{-1}A) = n - \operatorname{Tr}\big(K_\epsilon^{-1}(C+\epsilon I)\big) \;\ge\; n - \operatorname{Tr}\big(C_\epsilon^{-1}(C+\epsilon I)\big) = \operatorname{Tr}(C_\epsilon^{-1}A_w),$$

where the inequality uses $C + \epsilon I \succeq 0$. Cauchy–Schwarz for the Hilbert–Schmidt inner product, applied to $C_\epsilon^{-1/2}A_w^{1/2}$ and $C_\epsilon^{1/2}A_w^{1/2}$, gives $(\operatorname{Tr} A_w)^2 \le \operatorname{Tr}(C_\epsilon^{-1}A_w)\operatorname{Tr}(C_\epsilon A_w)$. By the addition theorem $\operatorname{Tr} A_w = n\lambda D$ pathwise. Letting $\epsilon \downarrow 0$ (the Gram matrix is invertible almost surely),

$$T_{\le L} = \operatorname{Tr}(K^{-1}A) \;\ge\; \frac{(n\lambda D)^2}{\operatorname{Tr}(A_w^2) + \operatorname{Tr}(CA_w)}. \tag{6}$$

The denominator has exact expectations. By (1), for independent X, Y, $\mathbb{E}[\Pi_{\le L}(X,Y)^2] = \sum_{s\le L}\mathbb{E}_Y\Pi_s(X,Y)^2 = \sum_{s\le L}\Pi_s(X,X) = D$ where $\Pi_{\le L} = \sum_{s\le L}\Pi_s$, while $\Pi_{\le L}(X,X) = D$; hence $\mathbb{E}\operatorname{Tr}(A_w^2) = \lambda^2\big(nD^2 + n(n-1)D\big)$. Likewise $\mathbb{E}[C(X,Y)A_w(X,Y)] = 0$ for $X \ne Y$ because the two kernels expand over disjoint sets of degrees, while $C(X,X) = \tau_{>L}$ and $A_w(X,X) = \lambda D$; hence $\mathbb{E}\operatorname{Tr}(CA_w) = n\tau_{>L}\lambda D$. No independence between C and A_w is used here: both are functions of the same design, and the off-diagonal expectation vanishes by orthogonality alone. Since $z \mapsto 1/z$ is convex on $(0,\infty)$, Jensen's inequality applied to (6) gives

$$\mathbb{E}\, T_{\le L} \;\ge\; \frac{(n\lambda D)^2}{\lambda^2(nD^2 + n(n-1)D) + n\tau_{>L}\lambda D} = \frac{nD}{D + n - 1 + \tau_{>L}/\lambda}.$$ □

Lemma 4.4 (the last subcritical block)**.** *Let $n = 2^d$ and $L_- = L_-(d) = \max\{L : \sum_{s\le L} N_{d,s} \le n\}$, $D_- = \sum_{s\le L_-} N_{d,s}$ and $y_d = D_-/n \in (0,1]$. Let $c_* \in (0,1)$ be the root of $H(c) = \log 2$, where $H(c) = (1+c)\log(1+c) - c\log c$. Then $L_-/d \to c_*$, $c_* > 1/4$, and*

$$\liminf_{d\to\infty} y_d \;\ge\; \frac{c_*}{1+c_*} \;>\; \frac{1}{5}, \qquad \frac{\tau_{>L_-}}{n\,\lambda_{d,L_-}} \;\longrightarrow\; 0. \tag{7}$$

Proof. H is continuous and strictly increasing on $(0,\infty)$ ($H'(c) = \log\frac{1+c}{c} > 0$), with $H(0^+) = 0$ and $H(1/4) = \frac{5}{4}\log 5 - 2\log 2 < \log 2$ (numerically $0.6255 < 0.6931$), so c_* exists and $c_* > 1/4$. By Stirling's formula, $\frac{1}{d}\log\binom{d+cd}{cd} \to H(c)$ uniformly for c in compact subsets of $(0,\infty)$. Fix $c < c_*$: by Lemma 3.1, $\sum_{s\le cd} N_{d,s} \le (cd+1)\binom{d+cd}{cd} = e^{d(H(c)+o(1))} < 2^d$ for large d, so $L_- \ge cd$. Fix $c \in (c_*, 1)$: $N_{d,cd} \ge \frac{3}{4}\binom{d+cd-1}{cd} = e^{d(H(c)+o(1))} > 2^d$ for large d, so $L_- < cd$. Hence $L_-/d \to c_*$, and by Lemma 3.1 $\rho_{L_-} = \frac{(d+2L_-)(d+L_--2)}{(L_-+1)(d+2L_--2)} \to \frac{1+c_*}{c_*} =: r_*$, with $r_* < 5$ because $c_* > 1/4$.

For the first claim in (7): $n < D_- + N_{L_-+1}$ by maximality of L_-, so $y_d > \big(1 + N_{L_-+1}/D_-\big)^{-1}$. Fix an integer J. For $1 \le j \le J$ and d large, $\rho_{L_--j} = \rho_{L_-}\prod_{i=1}^{j}\frac{\rho_{L_--i}}{\rho_{L_--i+1}} \le \rho_{L_-}(1+1/(L_- - J))^{2J}$, because, from the product form of ρ_s in Lemma 3.1, $\frac{\rho_{s-1}}{\rho_s} = \frac{d+s-3}{d+s-2}\cdot\frac{s+1}{s}\cdot\frac{(d+2s-2)^2}{(d+2s)(d+2s-4)} \le \big(1+\frac{1}{s}\big)\big(1+\frac{4}{(d+2s)(d+2s-4)}\big) \le \big(1+\frac{1}{s}\big)^2$ for $s \ge 2$. Writing $\bar\rho = \rho_{L_-}(1+1/(L_- - J))^{2J}$, $D_- \ge \sum_{j=0}^{J} N_{L_--j} \ge N_{L_-}\sum_{j=0}^{J}\bar\rho^{-j}$, while $N_{L_-+1} = \rho_{L_-}N_{L_-}$. Since $L_- \to \infty$, $\bar\rho \to r_*$, and therefore $\liminf_d y_d \ge \big(1 + r_*/\sum_{j=0}^{J} r_*^{-j}\big)^{-1}$ for every J; letting $J \to \infty$, $\liminf_d y_d \ge \big(1 + r_*(1-1/r_*)\big)^{-1} = 1/r_* = c_*/(1+c_*) > 1/5$.

For the second claim, by Lemma 3.1, for $s \ge L_- \ge d/10$ (large d), $\rho_s \le 2\,\frac{d+s-2}{s+1} \le 22$, and by Lemma 3.2, $\lambda_{s+1}/\lambda_s \le \gamma/d$. Hence for $j \ge 1$, with $L = L_-$, $\frac{\tau_{L+j}}{n\lambda_L} = \frac{N_{L+j}}{n}\cdot\frac{\lambda_{L+j}}{\lambda_L} \le \frac{N_L}{n}\,(22\gamma/d)^j \le (22\gamma/d)^j$, using $N_L \le D_- \le n$. Summing over $j \ge 1$ gives $\tau_{>L}/(n\lambda_L) \le 44\gamma/d$ for $d > 44\gamma$. □

Proof of Theorem 4.1. Since $\gamma = h^{-2}$ is fixed, $d > \gamma$ for all large d, so Lemmas 3.2 and 4.3 apply. Take $B = \{0,\dots,L_-\}$ in Lemma 4.2, so $D_B = D_-$ and $T_B = T_{\le L_-}$. Taking expectations in (4) and applying Jensen's inequality, $\mathbb{E}[T^2] \ge (\mathbb{E}T)^2$, then Lemma 4.3,

$$\frac{\mathbb{E}\,\mathcal{E}_{2^d,d}}{\sigma^2} = \mathbb{E}\frac{V(X)}{\sigma^2} \;\ge\; \frac{(\mathbb{E}T_{\le L_-})^2}{nD_-} \;\ge\; \frac{nD_-}{\big(D_- + n - 1 + \tau_{>L_-}/\lambda_{d,L_-}\big)^2} = \frac{y_d}{\big(1 + y_d - \frac{1}{n} + \eta_d\big)^2},$$

with $\eta_d = \tau_{>L_-}/(n\lambda_{d,L_-}) \to 0$ by Lemma 4.4. The function $g(y) = y/(1+y)^2$ is increasing on $(0,1]$, since $g'(y) = (1-y)/(1+y)^3 \ge 0$. Given $\epsilon > 0$, Lemma 4.4 gives $y_d \ge \frac{1}{5} - \epsilon$ for all large d, and $y_d \le 1$ always, so for large d the right side is at least $g(\frac{1}{5} - \epsilon) - o(1)$. Letting $d \to \infty$ and then $\epsilon \to 0$, $\liminf_d \mathbb{E}\,\mathcal{E}_{2^d,d}/\sigma^2 \ge g(1/5) = \frac{1/5}{(6/5)^2} = \frac{5}{36}$. □

Remark 4.5. The argument gives more than the stated constant. Lemma 4.4 shows $\liminf y_d \ge 1/r_*$ with $1/r_* = c_*/(1+c_*) = 0.2271\ldots$, so the same proof yields the constant $g(1/r_*) = 0.1509\ldots$; the constant $5/36$ is the value at the round number $1/5$. The bound is an expectation statement from start to finish: no concentration of the leverage is used, and none is needed. The certificate that the proof evaluates, $(T_{\le L}/n)^2/(D_{\le L}/n)$, and the block dimension ratio $D_{\le L}/n$ are exactly the quantities measured in Section 7, where the one-window bound is compared with the realised $V(X)/\sigma^2$ at $d \le 13$. Running the same argument over several blocks simultaneously improves the constant; the ten-window certificate $c_{10} = 3882594969529/12500000000000 = 0.3106\ldots$ obtained in [45] is the value against which the measurements of Section 7 are also reported.

5 Target 2: benign overfitting along a doubly exponential sequence

Here $\ell \geq 1$ is the level, $R = 2^\ell$, $d = 2^R$ and $n = d^R$, so that $d_\ell = 2^{2^\ell}$ and $n_\ell = 2^{2^{2\ell}}$. The labels are $Y_i = f_d(X_i) + \varepsilon_i$ with $\varepsilon \sim N(0, \sigma^2 I_n)$ independent of the design, where $f_d \in \bigoplus_{s \leq q_0} \mathcal{H}_{d,s}$ for a degree bound q_0 that does not depend on ℓ, and $\|f_d\|_{L^2(S^{d-1})} \leq B$. Let $\hat{f}$ be the interpolant of the data of minimal norm in the reproducing kernel Hilbert space $\mathcal{H}_K$ of K, and

$$\mathcal{R}_\ell = \|\hat{f} - f_d\|^2_{L^2(S^{d-1})} = \mathbb{E}_{X_0}\big[(\hat{f}(X_0) - f_d(X_0))^2 \,\big|\, X, \varepsilon\big].$$

Theorem 5.1 (Target 2). *For every fixed $h > 0$, q_0, B and σ, and every $\epsilon > 0$, $\sum_\ell \Pr\{\mathcal{R}_\ell > \epsilon\} < \infty$. In particular $\mathcal{R}_\ell \to 0$ in probability, and almost surely along the sequence under any coupling of the levels.*

The critical degree is $k = R + \ell - 1$; degrees $s < k$ form the low block, of dimension $D = D_{<k} = \sum_{s<k} N_s$, and degrees $s \geq k$ form the tail. We write $\delta_* = \frac{1}{\log 2} - 1 = 0.4426\ldots$, and throughout "for large ℓ" means for all ℓ beyond a threshold depending only on γ.

Lemma 5.2 (scales at the critical degree). *As $\ell \to \infty$,*

$$\frac{N_k}{n} = d^{\,\delta_* + o(1)}, \qquad \frac{D}{n} = d^{-(1-\delta_*)+o(1)},$$

while $\log n = R^2 \log 2$, $n^{1/R} = d$, and N_s is non-decreasing in s.

Proof. Since $k^2 = O(R^2) = o(d)$, Lemma 3.1 gives $\log N_k = k \log d - \log k! + o(1)$. With $\log d = R \log 2$, $\ell = \log_2 R$ and Stirling's formula $\log k! = k \log k - k + O(\log k)$,

$$\begin{aligned} \log \frac{N_k}{n} &= (\ell - 1) R \log 2 - (R + \ell - 1) \log(R + \ell - 1) + (R + \ell - 1) + O(\log R) \\ &= R(1 - \log 2) - (\ell - 1) \log R + O(\ell), \end{aligned}$$

because $(\ell - 1) \log 2 = \log(R/2)$ and $\log(R + \ell - 1) = \log R + O(\ell/R)$. Dividing by $\log d = R \log 2$ gives $\log(N_k/n)/\log d \to (1 - \log 2)/\log 2 = \delta_*$. For the low block, Lemma 3.1 gives $1/\rho_s \leq \frac{s+1}{d+s-2} \leq \frac{2k}{d}$ for $s < k$ and $d \geq 4k$, hence $N_{k-j} \leq N_k (2k/d)^j$ and $D \leq N_k \sum_{j \geq 1} (2k/d)^j \leq 4k N_k/d$, so $D/n \leq d^{\delta_* - 1 + o(1)}$; the matching lower bound $D \geq N_{k-1} \geq N_k\, d^{-1+o(1)}$ follows from $\rho_{k-1} \leq \frac{d+2k}{k} \cdot \frac{d+k}{d+2k-2} \leq d^{1+o(1)}$. The identities for $\log n$ and $n^{1/R}$ are immediate from $n = d^R$, $d = 2^R$, and monotonicity of N_s is $\rho_s \geq 1$ in Lemma 3.1. □

Lemma 5.3 (the tail kernel). *Let $\tau = \sum_{s \geq k} \tau_s$, $w_s = \tau_s/\tau$ for $s \geq k$, $K_T = \sum_{s \geq k} \lambda_s \Pi_s$ and $r_T = K_T/\tau = \sum_{s \geq k} w_s r_s$ with $r_s = \Pi_s / N_s$. Then $r_T(x, x) = 1$, $\mathbb{E}_Y r_T(x, Y) = 0$, and for large ℓ:*

(i) *$\tau_{k+r}/\tau_k \leq 2 e^\gamma \gamma^r k!/(k + r)!$ for all $r \geq 0$; hence $1 - w_k \leq 4 e^\gamma \gamma/(k + 1)$;*

(ii) *for every $z \in S^{d-1}$ and $1 \leq q \leq R$, $\|r_T(X, z)\|_{L^{2q}} \leq 4 e^\gamma (2q - 1)^{k/2} N_k^{-1/2}$;*

(iii) *$\mathbb{E}\, r_T(X, z)^2 = \sum_{s \geq k} w_s^2/N_s \in \big[w_k^2/N_k,\, 1/N_k\big]$, so $\mathbb{E}\, r_T(X, z)^2 = (1 + o(1))/N_k$;*

(iv) *with $G_T = (\mathbb{E}_z[r_T(X_i, z) r_T(z, X_j)])_{ij}$ and $R_T = (r_T(X_i, X_j))_{ij}$, $0 \preceq G_T \preceq N_k^{-1} R_T$ and $(G_T)_{jj} \leq 1/N_k$.*

Proof. $r_T(x,x)=\sum w_s=1$ and $\mathbb{E}_Y r_T(x,Y)=0$ follow from (1) since every tail degree is positive. (i) By (3), $\tau_{k+r}\le e^{-\gamma}\sum_{m\ge k+r}\gamma^m/m!\le\gamma^{k+r}/(k+r)!$, while $\tau_k=N_k\lambda_k\ge N_k e^{-\gamma}\gamma^k/(d)_k\ge\frac12 e^{-\gamma}\gamma^k/k!$ for large ℓ, because $N_k k!/(d)_k=e^{O(k^2/d)}\prod_{j<k}(1+2j/d)^{-1}\to1$ by Lemma 3.1. The bound on $1-w_k$ follows from $\sum_{r\ge1}\gamma^r k!/(k+r)!\le\sum_{r\ge1}(\gamma/(k+1))^r\le 2\gamma/(k+1)$. (ii) By the triangle inequality, Lemma 3.3, monotonicity of N_s and $w_{k+r}\le\tau_{k+r}/\tau_k$,

$$\|r_T(X,z)\|_{2q}\le\sum_{r\ge0}w_{k+r}\frac{(2q-1)^{(k+r)/2}}{\sqrt{N_{k+r}}}\le\frac{(2q-1)^{k/2}}{\sqrt{N_k}}\,2e^{\gamma}\sum_{r\ge0}\Big(\frac{\gamma\sqrt{2q-1}}{k+1}\Big)^r\le\frac{4e^{\gamma}(2q-1)^{k/2}}{\sqrt{N_k}},$$

since $\gamma\sqrt{2R}/(k+1)\le1/2$ for large ℓ. (iii) By (1), $\mathbb{E}[r_s(X,z)r_t(X,z)]=\delta_{st}/N_s$; the bounds use $\sum_s w_s^2\le(\sum_s w_s)^2=1$ and $N_s\ge N_k$. (iv) By (1), $\mathbb{E}_z[r_s(X_i,z)r_t(z,X_j)]=\delta_{st}r_s(X_i,X_j)/N_s$, so $G_T=\sum_{s\ge k}(w_s/N_s)\,w_s r_s(X,X)$; each $w_s r_s(X,X)$ is positive semidefinite and $w_s/N_s\le1/N_k$, which gives both claims. □

Lemma 5.4 (the row-square statistic)**.** *Let $\widetilde X$ be an independent copy of the design and, for $1\le j\le n$, $S_j=\sum_{i\ne j}r_T(X_i,\widetilde X_j)^2$. Then for large ℓ,*

$$\mathbb{E}\max_{1\le j\le n}S_j\;\le\;(1+o(1))\,\frac{n}{N_k}.$$

Proof. Fix $z\in S^{d-1}$ and put $Y_i=r_T(X_i,z)^2-\mathbb{E}r_T(X_i,z)^2$, independent, centred, and identically distributed. Rosenthal's inequality with the constant of Johnson, Schechtman and Zinn [47, 30] gives, for $q\ge2$, $\|\sum_{i\ne j}Y_i\|_q\le C_0 q\big(\sqrt n\,\|Y_1\|_2+n^{1/q}\|Y_1\|_q\big)$ with an absolute constant C_0. Take $q=R$. By Lemma 5.3(ii) with $q=2$ and $q=R$, $\|Y_1\|_2\le\|r_T\|_4^2\le C_\gamma 3^k/N_k$ and $\|Y_1\|_R\le 2\|r_T\|_{2R}^2\le C_\gamma(2R)^k/N_k$, and $n^{1/R}=d$. The bounds do not depend on z, so

$$\mathbb{E}\max_j\big|S_j-\mathbb{E}S_j\big|\le\Big(\sum_j\mathbb{E}|S_j-\mathbb{E}S_j|^R\Big)^{1/R}\le n^{1/R}\sup_z\big\|\textstyle\sum_{i\ne j}Y_i\big\|_R\le C_\gamma\,dR\Big(\frac{\sqrt n\,3^k}{N_k}+\frac{d\,(2R)^k}{N_k}\Big).$$

Relative to $\mathbb{E}S_j\le n/N_k$, the two terms are $dR3^k/\sqrt n$ and $Rd^2(2R)^k/n$. In base d, using $k=R+\ell-1$, $\log d=R\log 2$ and $2R=2^{\ell+1}$: $3^k=d^{(\log_2 3)(1+(\ell-1)/R)}$ and $(2R)^k=d^{(\ell+1)(1+(\ell-1)/R)}$, so the two ratios are $R\,d^{\,1+\log_2 3+o(1)-R/2}$ and $R\,d^{\,3+\ell+o(\ell)-R}$, both of which tend to zero because $R=2^\ell$. Since $\mathbb{E}S_j=(n-1)\mathbb{E}r_T(X,z)^2\le n/N_k$ by Lemma 5.3(iii), the claim follows. □

Proposition 5.5 (isotropy of the tail Gram matrix)**.** *Let $R_T=(r_T(X_i,X_j))_{ij}$ and $A_d=\mathbb{E}\|R_T-I\|_{\mathrm{op}}$. Then $A_d\to0$; more precisely, for every fixed $\eta<\delta_*$ and large ℓ, $A_d\le C_{\gamma,\eta}R^2d^{-\eta/2}$.*

The proof follows the method of Kaushik, Romberg and Muthukumar [31]: decoupling, then the noncommutative Khintchine inequality. It is written out because the quantity that must be controlled at the growing degree k is the row-square statistic of Lemma 5.4, and the maximum-entry relaxation of that statistic which appears in the statement of [31, Theorem 1] is too large here by a factor of order d^ℓ.

Proof. Since $r_T(x,x)=1$, $\Delta:=R_T-I=\sum_{i\ne j}r_T(X_i,X_j)e_ie_j^\top$ is a matrix-valued U-statistic of order two. By the decoupling inequality for U-statistics with values in a Banach space [16, 15], applied to the convex function $\|\cdot\|_{\mathrm{op}}$, there is an absolute constant C_1 with

$$\mathbb{E}\|\Delta\|\le C_1\mathbb{E}\|\widetilde\Delta\|,\qquad\widetilde\Delta=\sum_{i\ne j}r_T(X_i,\widetilde X_j)e_ie_j^\top=\sum_{j=1}^n c_je_j^\top,\qquad c_j=\big(r_T(X_i,\widetilde X_j)\mathbf1_{\{i\ne j\}}\big)_{i\le n},$$

where $\widetilde{X}$ is an independent copy of X. Condition on X. The columns $c_1, \dots, c_n$ are then independent (each is a function of its own $\widetilde{X}_j$) and centred, by Lemma 5.3. Moreover $\mathbb{E}[c_j c_j^\top \mid X] = P_j G_T P_j =: M_j$ with $P_j = I - e_j e_j^\top$ and G_T as in Lemma 5.3(iv); hence $\|M_j\| \le \|G_T\|$ and

$$\Big\| \sum_j M_j \Big\| = \big\| (n-2) G_T + \operatorname{diag}(G_T) \big\| \le n \|G_T\| + \frac{1}{N_k}, \qquad \Big\| \sum_j M_j^2 \Big\| \le n \|G_T\|^2. \tag{8}$$

All expectations below are conditional on X unless stated otherwise. Write $\|c_j\|^2 = S_j$ as in Lemma 5.4, $m_2 = \mathbb{E} \max_j S_j$, and $U = \mathbb{E} \| \sum_j c_j c_j^\top \|$; thus m_2 and U are X-measurable random variables, and it is their X-expectations that enter at the end of the proof, where Lemma 5.4 bounds $\mathbb{E} m_2$.

First application of Khintchine. By symmetrisation [34, Lemma 6.3] and the matrix Khintchine inequality for Rademacher series [55, Theorem 4.1.1], applied to the $n \times n$ matrices $c_j e_j^\top$, for which

$$\sum_j (c_j e_j^\top)(c_j e_j^\top)^\top = \sum_j c_j c_j^\top, \qquad \sum_j (c_j e_j^\top)^\top (c_j e_j^\top) = \operatorname{diag}(S_1, \dots, S_n),$$

we get

$$\mathbb{E} \|\widetilde{\Delta}\| \le 2\sqrt{2 \log(2n)}\, \mathbb{E} \max \Big(\big\| \textstyle\sum_j c_j c_j^\top \big\|^{1/2},\ \max_j \sqrt{S_j} \Big) \le 2\sqrt{2 \log(2n)} \left(\sqrt{U} + \sqrt{m_2} \right), \tag{9}$$

by Jensen's inequality.

Second application. The matrices $Z_j = c_j c_j^\top - M_j$ are independent, centred and symmetric, so by the same two inequalities, $\mathbb{E} \| \sum_j Z_j \| \le 2\sqrt{2 \log(2n)}\, \mathbb{E} \| \sum_j Z_j^2 \|^{1/2}$. Since $Z_j^2 \preceq 2(c_j c_j^\top)^2 + 2M_j^2 = 2 S_j c_j c_j^\top + 2M_j^2$, (8) gives $\| \sum_j Z_j^2 \| \le 2 \max_j S_j \, \| \sum_j c_j c_j^\top \| + 2n \|G_T\|^2$, and therefore, by Cauchy–Schwarz,

$$U \le \Big\| \sum_j M_j \Big\| + \mathbb{E} \Big\| \sum_j Z_j \Big\| \le n \|G_T\| + \frac{1}{N_k} + 4\sqrt{\log(2n)} \Big(\sqrt{m_2 U} + \sqrt{n}\, \|G_T\| \Big).$$

Using $4\sqrt{\log(2n)} \sqrt{m_2 U} \le U/2 + 8 \log(2n)\, m_2$ and rearranging,

$$U \le 2n \|G_T\| \Big(1 + 4\sqrt{\tfrac{\log(2n)}{n}} \Big) + \frac{2}{N_k} + 16 \log(2n)\, m_2. \tag{10}$$

Closing the recurrence. Take expectations over X. By Lemma 5.3(iv), $\mathbb{E} \|G_T\| \le \mathbb{E} \|R_T\| / N_k \le (1 + A_d)/N_k$, and by Lemma 5.4, $\mathbb{E} m_2 \le (1 + o(1)) n / N_k$. Inserting (10) into (9) (with $\mathbb{E}\sqrt{U} \le \sqrt{\mathbb{E} U}$) and then into the decoupling bound,

$$A_d \le C \sqrt{\log n} \Big(\sqrt{\frac{n(1 + A_d)}{N_k}} + \sqrt{\frac{\log n \ n}{N_k}} \Big) \le C \sqrt{\frac{n \log n}{N_k}} \left(\sqrt{1 + A_d} + \sqrt{\log n} \right)$$

for large ℓ, with C absolute. Put $\epsilon_d = n \log n / N_k$; by Lemma 5.2, $\epsilon_d \log n = d^{-\delta_* + o(1)} R^4 \log^2 2 \to 0$. If $A_d \ge 1$ the inequality gives

$$\sqrt{A_d} \le C \sqrt{\epsilon_d} \Big(\sqrt{2} + \frac{\sqrt{\log n}}{\sqrt{A_d}} \Big) \le C \sqrt{\epsilon_d} \big(\sqrt{2} + \sqrt{\log n} \big),$$

which is smaller than one for large ℓ, a contradiction; so $A_d < 1$ for large ℓ, and then

$$A_d \le C \sqrt{\epsilon_d} \big(\sqrt{2} + \sqrt{\log n} \big) \le C' \sqrt{\epsilon_d \log n} = C' R^2 d^{-\delta_*/2 + o(1)},$$

which is at most $C_{\gamma,\eta} R^2 d^{-\eta/2}$ for every fixed $\eta < \delta_*$. □

Lemma 5.6 (the interpolant through the low block)**.** *Let $\Psi \in \mathbb{R}^{n\times D}$ be the matrix of an L^2-orthonormal basis of $\bigoplus_{s<k}\mathcal{H}_{d,s}$ evaluated at the design, Λ the diagonal matrix of the corresponding eigenvalues λ_s, and $T = (K_T(X_i,X_j))_{ij} = \tau R_T$. Let $E_1 = \{\|\Psi^\top\Psi/n - I_D\| \le 1/2\}$ and $E_2 = \{\|R_T - I\| \le 1/2\}$. Then $\Pr(E_1^c) \le 2De^{-n/(10D)}$ and $\Pr(E_2^c) \le 2A_d$. On $E_1\cap E_2$ the minimal-norm interpolant is $\hat f = \Psi(\cdot)^\top\hat a + k_T(\cdot)^\top T^{-1}(y-\Psi\hat a)$ with $k_T(x) = (K_T(x,X_i))_i$ and*

$$\hat a = G_0^{-1}\Psi^\top W y, \qquad W = \tau T^{-1}, \qquad G_0 = \tau\Lambda^{-1} + \Psi^\top W\Psi, \qquad \tfrac{2}{3}I \preceq W \preceq 2I, \qquad \|G_0^{-1}\| \le \frac{3}{n}. \tag{11}$$

Proof. The rows $\Psi(X_i)$ are independent, and by (1)

$$\|\Psi(X_i)\|^2 = \sum_{s<k}\Pi_s(X_i,X_i) = D, \qquad \mathbb{E}\,\Psi(X_i)\Psi(X_i)^\top = I_D.$$

The matrix Chernoff inequality of Tropp [54, Theorem 1.1], at deviation $1/2$, gives

$$\Pr(E_1^c) \le D\big(e^{-1/2}\sqrt{2}\big)^{n/D} + D\big(e^{1/2}(3/2)^{-3/2}\big)^{n/D} \le 2De^{-n/(10D)}.$$

Markov's inequality gives $\Pr(E_2^c) \le 2A_d$. The RKHS $\mathcal{H}_K$ is the orthogonal sum of $\mathcal{H}_{\mathrm{low}} = \bigoplus_{s<k}\mathcal{H}_{d,s}$ with norm $\|\Psi(\cdot)^\top a\|^2_{\mathcal{H}_K} = a^\top\Lambda^{-1}a$ and of the tail space $\mathcal{H}_T$ with kernel K_T. For fixed low coefficients a, the tail function of minimal norm interpolating $y-\Psi a$ at the design is $k_T(\cdot)^\top T^{-1}(y-\Psi a)$, of squared norm $(y-\Psi a)^\top T^{-1}(y-\Psi a)$ (T is invertible on E_2). The minimal-norm interpolant therefore minimises $J(a) = a^\top\Lambda^{-1}a + (y-\Psi a)^\top T^{-1}(y-\Psi a)$, a strictly convex quadratic whose stationarity condition is $(\Lambda^{-1}+\Psi^\top T^{-1}\Psi)a = \Psi^\top T^{-1}y$; multiplying by τ gives (11). On E_2, $\frac12 I \preceq T/\tau \preceq \frac32 I$, hence $\frac23 I \preceq W \preceq 2I$; on $E_1\cap E_2$, $\Psi^\top W\Psi \succeq \frac23\cdot\frac{n}{2}I = \frac{n}{3}I$, so $G_0 \succeq \frac{n}{3}I$. □

Proof of Theorem 5.1. Take ℓ large enough that $q_0 < k$ and write $f_d = \Psi(\cdot)^\top\theta$ with $\|\theta\| = \|f_d\|_{L^2} \le B$ and θ supported on degrees $\le q_0$. Work on $E_1\cap E_2$. Since low and tail functions are orthogonal in $L^2(S^{d-1})$ and $\|\Psi(\cdot)^\top a\|_{L^2} = \|a\|$,

$$\mathcal{R}_\ell = \|\hat a - \theta\|^2 + \|f_T\|^2_{L^2}, \qquad f_T = k_T(\cdot)^\top T^{-1}r, \qquad r = y - \Psi\hat a.$$

The low coefficients. With $y = \Psi\theta+\varepsilon$ and (11), $\hat a-\theta = -\tau G_0^{-1}\Lambda^{-1}\theta + G_0^{-1}\Psi^\top W\varepsilon$. For the first term, $\tau\le 1$ and, by (3), $\lambda_{d,s} \ge e^{-\gamma}\gamma^s/(d)_s \ge c_{\gamma,q_0}d^{-q_0}$ for $s\le q_0$, so $\|\tau G_0^{-1}\Lambda^{-1}\theta\| \le \frac{3}{n}\cdot\frac{B}{c_{\gamma,q_0}}\,d^{q_0}$. For the second, $\mathbb{E}_\varepsilon\|G_0^{-1}\Psi^\top W\varepsilon\|^2 = \sigma^2\operatorname{Tr}\big(G_0^{-1}\Psi^\top W^2\Psi G_0^{-1}\big) \le \sigma^2\|G_0^{-1}\|^2\|W\|^2\operatorname{Tr}(\Psi^\top\Psi) \le \sigma^2\cdot\frac{9}{n^2}\cdot 4\cdot nD = \frac{36\sigma^2 D}{n}$. Hence

$$\mathbb{E}_\varepsilon\|\hat a-\theta\|^2 \le C_{\gamma,q_0}\frac{B^2d^{2q_0}}{n^2} + \frac{72\sigma^2 D}{n}. \tag{12}$$

The tail function. By (1), $\|f_T\|^2_{L^2} = r^\top T^{-1}H_T T^{-1}r$ with

$$H_T = \mathbb{E}_{X_0}\big[k_T(X_0)k_T(X_0)^\top\big] = \sum_{s\ge k}\lambda_s^2\Pi_s(X,X) \preceq \lambda_k T,$$

because $\lambda_s \le \lambda_k$ for $s\ge k$ (Lemma 3.2). On E_2, $T\succeq\frac{\tau}{2}I$, so

$$\|f_T\|^2_{L^2} \le \lambda_k\, r^\top T^{-1}r \le \frac{2\lambda_k}{\tau}\|r\|^2 \le \frac{2}{N_k}\|r\|^2,$$

using $\lambda_k/\tau \le \lambda_k N_k/(\tau_k N_k) \le 1/N_k$. Since $r = \Psi(\theta - \hat{a}) + \varepsilon$ and $\|\Psi\|^2 \le \frac{3}{2}n$ on E_1, $\|r\|^2 \le 3n\|\theta - \hat{a}\|^2 + 2\|\varepsilon\|^2$, and with (12),

$$\mathbb{E}_\varepsilon \|f_T\|_{L^2}^2 \le \frac{6n}{N_k}\Big(C_{\gamma,q_0}\frac{B^2 d^{2q_0}}{n^2} + \frac{72\sigma^2 D}{n}\Big) + \frac{4n\sigma^2}{N_k}.$$

The conditional risk bound. Combining the two displays, on $E_1 \cap E_2$,

$$\mathbb{E}_\varepsilon[\mathcal{R}_\ell \mid X] \le C_{\gamma,q_0}\Big[\sigma^2\Big(\frac{D}{n} + \frac{n}{N_k}\Big) + B^2\Big(\frac{d^{2q_0}}{n^2} + \frac{d^{2q_0}}{nN_k}\Big)\Big] = C_{\gamma,q_0,B,\sigma}\, d^{-\delta_* + o(1)}, \tag{13}$$

where the last step is Lemma 5.2: $n/N_k = d^{-\delta_* + o(1)}$ is the slowest term, since $D/n = d^{-(1-\delta_*)+o(1)}$ with $1 - \delta_* > \delta_*$, and $d^{2q_0}/n = d^{2q_0 - R}$.

Conclusion. By Markov's inequality in the noise, $\Pr\{\mathcal{R}_\ell > \epsilon \mid X\} \le \epsilon^{-1}\mathbb{E}_\varepsilon[\mathcal{R}_\ell \mid X]$ on $E_1 \cap E_2$, so by Lemma 5.6 and Proposition 5.5,

$$\Pr\{\mathcal{R}_\ell > \epsilon\} \le 2De^{-n/(10D)} + 2C_{\gamma,\eta}R^2 d^{-\eta/2} + \frac{C_{\gamma,q_0,B,\sigma}}{\epsilon}\, d^{-\delta_* + o(1)}.$$

Every term is summable in ℓ because $d = d_\ell = 2^{2^\ell}$ and $n/D \ge d^{1-\delta_* + o(1)}$. This proves the summability claim; convergence in probability follows, and the first Borel–Cantelli lemma gives $\mathcal{R}_\ell \to 0$ almost surely whatever the joint law of the levels, since only the marginal tail probabilities enter. □

Remark 5.7. The mechanism is a dimension gap: the tail Gram matrix is isotropic because $n \ll N_k$, and the low block is well conditioned because $D \ll n$; both gaps are polynomial in d with the exponents δ_* and $1 - \delta_*$ of Lemma 5.2. Section 8 measures the isotropy of R_T and the risk of the interpolant as functions of the finite-size analogue $\delta = \log(N_{d,k}/n)/\log d$ of these exponents, on both sides of $\delta = 0$, which is where Theorem 4.1 lives.

6 Target 1: the correlated Gaussian equivalence

6.1 Model, surrogate and statement

Let $F = [f_1, \dots, f_m] \in \mathbb{R}^{d\times m}$ have independent $N(0, 1/d)$ entries. For each original sample $i \le n$ let a_i, z_{i1}, z_{i2} be independent $N(0, I_d)$ vectors, independent of F, and let $t_i = \xi^\top a_i$ with $\|\xi\| = 1$ fixed. The two augmented feature rows of sample i are

$$x_{ij}^A = \sigma\big(F^\top(a_i + \Delta z_{ij})\big) \in \mathbb{R}^m, \qquad j = 1, 2,$$

with $\Delta > 0$ fixed and $\sigma : \mathbb{R} \to \mathbb{R}$ odd, bounded, three times differentiable with bounded derivatives (for instance tanh). With $N = 2n$, $y \in \mathbb{R}^N$ the vector of the labels t_i each repeated twice and $X_A \in \mathbb{R}^{N\times m}$ the matrix of the rows x_{ij}^A, the objective is

$$\begin{aligned} L_A(w) &= \frac{1}{2N}\|y - X_A w\|^2 + \frac{\lambda}{2}\|w\|^2 \\ &= \frac{1}{4n}\sum_{i=1}^{n}\sum_{j=1}^{2}\big(t_i - (x_{ij}^A)^\top w\big)^2 + \frac{\lambda}{2}\|w\|^2, \end{aligned} \tag{14}$$

$\lambda > 0$ fixed, with minimiser $w_A^\star$ and minimum $L_A^\star$; $n/d \to \alpha > 0$ and $m/d \to \beta > 0$. A clean test point is $x^{0,A} = \sigma(F^\top a^0)$ with $t^0 = \xi^\top a^0$, $a^0 \sim N(0, I_d)$ independent of everything, and the population clean risk of a vector w is $R_A(w \mid F) = \mathbb{E}[(t^0 - (x^{0,A})^\top w)^2 \mid F, w]$.

The surrogate replaces σ by its first Hermite coefficients and the rest by independent Gaussian residuals. Let $s = \sqrt{1+\Delta^2}$, $\rho = s^{-2}$, let h_k be the orthonormal probabilists' Hermite polynomials, $G \sim N(0,1)$, and for $q > 0$

$$\begin{aligned} a_k(q) &= \mathbb{E}\big[\sigma(s\sqrt{q}\,G) h_k(G)\big], \qquad \kappa(q) = \mathbb{E}\big[\sigma'(s\sqrt{q}\,G)\big], \\ \eta(q) &= \sum_{k \geq 3 \text{ odd}} a_k(q)^2, \qquad \gamma(q) = \sum_{k \geq 3 \text{ odd}} a_k(q)^2 \rho^k. \end{aligned} \tag{15}$$

Since σ is odd, $a_k(q) = 0$ for even k, and Gaussian integration by parts gives $a_1(q) = s\sqrt{q}\,\kappa(q)$. Put $\mu_e = \kappa(1)$, $\mu_b = \Delta\kappa(1)$, $\mu_2 = \sqrt{\gamma(1)}$, $\mu_3 = \sqrt{\eta(1) - \gamma(1)}$ (both radicands are non-negative, see Lemma 6.3), and define the surrogate rows and clean row

$$x_{ij}^G = \mu_e F^\top a_i + \mu_b F^\top z_{ij} + \mu_2 b_i + \mu_3 p_{ij}, \qquad x^{0,G} = \nu_1 F^\top a^0 + \nu_2 g^0, \tag{16}$$

with $b_i, p_{i1}, p_{i2}, g^0 \sim N(0, I_m)$ independent of everything else, $\nu_1 = \mathbb{E}\sigma'(G)$ and $\nu_2^2 = \mathbb{E}\sigma(G)^2 - \nu_1^2$. The same a_i and the same b_i enter both views of sample i: the surrogate keeps the shared latent structure of the augmentation. The four training coefficients are those of Dhifallah and Lu [17]. L_G, $w_G^\star$, $L_G^\star$ and $R_G(w \mid F)$ are defined from (14) with X_G in place of X_A and $x^{0,G}$ in place of $x^{0,A}$.

Theorem 6.1 (Target 1)**.** *There are constants $C, c > 0$ depending only on $\sigma, \Delta, \lambda, \alpha, \beta$ such that for every $\varphi \in C_b^2(\mathbb{R})$ and all large d,*

$$\big|\mathbb{E}\varphi(L_A^\star) - \mathbb{E}\varphi(L_G^\star)\big| \leq C\Big(\|\varphi'\|_\infty \sqrt{\tfrac{\log d}{d}} + \|\varphi''\|_\infty \tfrac{1}{d}\Big) + C\|\varphi\|_\infty e^{-cd}, \tag{17}$$

$$\big|\mathbb{E}\varphi\big(R_A(w_A^\star \mid F)\big) - \mathbb{E}\varphi\big(R_G(w_G^\star \mid F)\big)\big| \leq C\Big(\|\varphi'\|_\infty \sqrt{\tfrac{\log d}{d}} + \|\varphi''\|_\infty \tfrac{1}{d}\Big) + C\|\varphi\|_\infty e^{-cd}. \tag{18}$$

In particular the optimised objective and the clean prediction risk of the augmented ridge problem have the same limits in law as those of the surrogate.

The proof replaces the n blocks (t_i, x_{i1}, x_{i2}) one at a time, in the manner of Lindeberg. Two exact facts make each step cheap: the change of the optimum under the insertion of one block is an explicit 2×2 quadratic form (Lemma 6.2), and conditional on F there is a Gaussian block whose second moments match those of the actual block exactly (Proposition 6.4), so that the only mismatch between the actual block and the Gaussian one is the fluctuation of a 2×2 leverage matrix, and the only mismatch between that Gaussian block and the surrogate is the replacement of the random column norms $q_r = \|f_r\|^2$ by one.

6.2 Exact identities

Lemma 6.2 (insertion of one block)**.** *Let $X_- \in \mathbb{R}^{(N-2)\times m}$ and y_- be the data with the two rows of sample i removed, let $L_-(w) = \frac{1}{2N}\|y_- - X_- w\|^2 + \frac{\lambda}{2}\|w\|^2$ with minimiser w_- and minimum $L_-^\star$, and put*

$$H_- = \lambda I + \tfrac{1}{N} X_-^\top X_-, \qquad B = H_-^{-1}, \qquad X_i = \begin{pmatrix} x_{i1}^\top \\ x_{i2}^\top \end{pmatrix}, \qquad e_i = t_i \mathbf{1}_2 - X_i w_-, \qquad Q_i = \tfrac{1}{N} X_i B X_i^\top.$$

Then the minimum $L^\star$ of the full objective satisfies

$$L^\star - L_-^\star = \frac{1}{2N}\, e_i^\top (I_2 + Q_i)^{-1} e_i, \qquad w^\star - w_- = \frac{1}{N}\, B X_i^\top (I_2 + Q_i)^{-1} e_i, \tag{19}$$

with $Q_i \succeq 0$, $\|(I_2 + Q_i)^{-1}\| \le 1$ and $\|B\| \le \lambda^{-1}$.

Proof. L_- is a quadratic with Hessian H_-, so $L_-(w) = L_-^\star + \frac{1}{2}(w - w_-)^\top H_-(w - w_-)$, and the full objective is $L(w) = L_-(w) + \frac{1}{2N}\|e_i - X_i(w - w_-)\|^2$. Minimising over $u = w - w_-$ gives $u^\star = (H_- + \frac{1}{N}X_i^\top X_i)^{-1}\frac{1}{N}X_i^\top e_i$ and $L^\star - L_-^\star = \frac{1}{2N} e_i^\top \big(I - \frac{1}{N}X_i(H_- + \frac{1}{N}X_i^\top X_i)^{-1}X_i^\top\big)e_i$. The push-through identity $\frac{1}{N}X_i(H_- + \frac{1}{N}X_i^\top X_i)^{-1}X_i^\top = Q_i(I + Q_i)^{-1}$ gives the first formula in (19), and $(H_- + \frac{1}{N}X_i^\top X_i)^{-1}X_i^\top = BX_i^\top(I + Q_i)^{-1}$ the second. The remaining claims follow from $H_- \succeq \lambda I$. □

Lemma 6.3 (residual covariances)**.** *For $q > 0$ let $V(q) = \mathbb{E}\sigma(s\sqrt{q}\,G)^2 = \sum_{k\ \mathrm{odd}} a_k(q)^2$ and, for a standard bivariate Gaussian (H_1, H_2) with correlation ρ, $C(q) = \mathbb{E}[\sigma(s\sqrt{q}H_1)\sigma(s\sqrt{q}H_2)] = \sum_{k\ \mathrm{odd}} a_k(q)^2\rho^k$. Then $\eta(q) = V(q) - s^2 q\kappa(q)^2$, $\gamma(q) = C(q) - q\kappa(q)^2$, $0 \le \gamma(q) \le \eta(q) \le \|\sigma\|_\infty^2$, and $V, \kappa, C, \eta, \gamma$ are Lipschitz on $[\frac{1}{2}, \frac{3}{2}]$ with constants depending only on σ and Δ.*

Proof. The expansions of V and C are Parseval's identity and Mehler's formula [29, Chapter 1] for the Hermite expansion of $\sigma(s\sqrt{q}\,\cdot)$; the $k = 1$ terms are $a_1(q)^2 = s^2 q\kappa(q)^2$ and $a_1(q)^2\rho = q\kappa(q)^2$, which gives the two identities and, since $0 < \rho < 1$, the inequalities. For the Lipschitz bounds, differentiation under the Gaussian expectation is justified by the bounded derivatives of σ: with $b(q) = s\sqrt{q}$, $V'(q) = \frac{s}{\sqrt{q}}\mathbb{E}[G\sigma(b(q)G)\sigma'(b(q)G)]$, $\kappa'(q) = \frac{s}{2\sqrt{q}}\mathbb{E}[G\sigma''(b(q)G)]$ and $C'(q) = \frac{s}{2\sqrt{q}}\mathbb{E}[H_1\sigma'(b(q)H_1)\sigma(b(q)H_2) + H_2\sigma(b(q)H_1)\sigma'(b(q)H_2)]$, all bounded on $[\frac{1}{2}, \frac{3}{2}]$. □

Proposition 6.4 (exact Gaussianisation conditional on F)**.** *Let $Q_F = \mathrm{diag}(q_1, \dots, q_m)$, $q_r = \|f_r\|^2$, $G_F = F^\top F$, $\mathscr{R} = Q_F^{-1/2} G_F Q_F^{-1/2}$ (a correlation matrix), $D_F = \mathrm{diag}(\kappa(q_r))$, $A_k = \mathrm{diag}(a_k(q_r))$, and*

$$\Sigma_{\mathrm{com}} = \sum_{k\ge 3\ \mathrm{odd}} \rho^k A_k \mathscr{R}^{\circ k} A_k, \qquad \Sigma_{\mathrm{priv}} = \sum_{k\ge 3\ \mathrm{odd}} (1 - \rho^k) A_k \mathscr{R}^{\circ k} A_k,$$

where $\circ k$ is the k-fold Hadamard power. Both are positive semidefinite. Conditional on F let $b_i^\# \sim N(0, \Sigma_{\mathrm{com}})$ and $p_{i1}^\#, p_{i2}^\# \sim N(0, \Sigma_{\mathrm{priv}})$ be independent of each other and of (a_i, z_{i1}, z_{i2}), and put

$$x_{ij}^\# = D_F F^\top a_i + \Delta D_F F^\top z_{ij} + b_i^\# + p_{ij}^\#. \tag{20}$$

Then, conditional on F, for $M \in \{A, \#\}$ the vectors $(t_i, x_{i1}^M, x_{i2}^M)$ have the same means, the same covariances and the same cross-covariances:

$$\begin{gathered}\mathbb{E}[t_i x_{ij}^M \mid F] = D_F F^\top \xi, \qquad \mathrm{Cov}(x_{ij}^M \mid F) = s^2 D_F G_F D_F + \Sigma_{\mathrm{com}} + \Sigma_{\mathrm{priv}}, \\ \mathrm{Cov}(x_{i1}^M, x_{i2}^M \mid F) = D_F G_F D_F + \Sigma_{\mathrm{com}}.\end{gathered} \tag{21}$$

Likewise, with $\kappa_0(q) = \mathbb{E}\sigma'(\sqrt{q}\,G)$, $a_{0,k}(q) = \mathbb{E}[\sigma(\sqrt{q}\,G)h_k(G)]$, $D_{0,F} = \mathrm{diag}(\kappa_0(q_r))$, $A_{0,k} = \mathrm{diag}(a_{0,k}(q_r))$ and $\Sigma_0 = \sum_{k\ge 3\ \mathrm{odd}} A_{0,k}\mathscr{R}^{\circ k}A_{0,k} \succeq 0$, the clean row $x^{0,\#} = D_{0,F}F^\top a^0 + r^0$, $r^0 \sim N(0, \Sigma_0)$ independent of a^0, satisfies

$$\mathbb{E}[t^0 x^{0,\#} \mid F] = \mathbb{E}[t^0 x^{0,A} \mid F] = D_{0,F}F^\top \xi, \qquad \mathrm{Cov}(x^{0,\#} \mid F) = \mathrm{Cov}(x^{0,A} \mid F) = D_{0,F}G_F D_{0,F} + \Sigma_0\,;$$

consequently $R_A(w \mid F) = R_\#(w \mid F)$ for every w that is independent of a^0.

Proof. Fix F. For view j and coordinate r the standardised pre-activation $H_{jr} = f_r^\top(a_i + \Delta z_{ij})/(s\sqrt{q_r})$ is standard Gaussian, with $\mathrm{Cov}(H_{jr}, H_{js}) = \mathscr{R}_{rs}$ within a view and $\mathrm{Cov}(H_{1r}, H_{2s}) = \rho\mathscr{R}_{rs}$ across views. Mehler's formula gives, for the actual rows,

$$\mathrm{Cov}(x_{ij}^A \mid F) = \sum_{k \text{ odd}} A_k \mathscr{R}^{\circ k} A_k, \qquad \mathrm{Cov}(x_{i1}^A, x_{i2}^A \mid F) = \sum_{k \text{ odd}} \rho^k A_k \mathscr{R}^{\circ k} A_k,$$

and the $k = 1$ terms are $A_1 \mathscr{R} A_1 = s^2 D_F G_F D_F$ and $\rho A_1 \mathscr{R} A_1 = D_F G_F D_F$ by $a_1(q) = s\sqrt{q}\,\kappa(q)$ and $\mathscr{R}_{rs}\sqrt{q_r q_s} = (G_F)_{rs}$. Since $(t_i, f_r^\top(a_i + \Delta z_{ij}))$ is centred jointly Gaussian with covariance $\xi^\top f_r$, Stein's identity gives $\mathbb{E}[t_i \sigma(f_r^\top(a_i + \Delta z_{ij})) \mid F] = \xi^\top f_r\, \mathbb{E}[\sigma'(f_r^\top(a_i + \Delta z_{ij})) \mid F] = \kappa(q_r)\xi^\top f_r$, which is $D_F F^\top \xi$. Positive semidefiniteness of Σ_{com} and Σ_{priv} follows from the Schur product theorem ($\mathscr{R}^{\circ k} \succeq 0$) and congruence by the diagonal matrices A_k. The moments of $x^\#$ are read off (20): the Gaussian block is centred, its teacher cross-covariance is $D_F F^\top \mathbb{E}[t_i a_i] = D_F F^\top \xi$, its within-view covariance is $D_F F^\top F D_F + \Delta^2 D_F F^\top F D_F + \Sigma_{\mathrm{com}} + \Sigma_{\mathrm{priv}}$ and its cross-view covariance drops the private terms; $1 + \Delta^2 = s^2$ gives (21). The clean statements are the same computation with $\Delta = 0$, and the risk identity holds because $R_M(w \mid F) = 1 - 2w^\top \mathbb{E}[t^0 x^{0,M} \mid F] + w^\top \mathrm{Cov}(x^{0,M} \mid F) w$ depends only on the matched moments. □

6.3 Estimates

Let Ω_F be the event that $\|F\|_{\mathrm{op}} \le 3(1 + \sqrt{\beta})$ and $\max_r |q_r - 1| \le \frac{1}{2}$. Since F has independent Gaussian entries of variance $1/d$ and $q_r \sim \chi_d^2/d$, $\Pr(\Omega_F^c) \le Ce^{-cd}$ [56, Theorem 4.4.5 and Section 2.8]. Constants C below depend only on $\sigma, \Delta, \lambda, \alpha, \beta$, and all statements are for large d.

Lemma 6.5 (a priori bounds)**.** *For every hybrid sample in which each block is either actual, of type #, or of type G, and every minimiser w of the corresponding objective (full or with one block removed), $\lambda\|w\|^2 \le \frac{1}{n}\sum_i t_i^2$; hence $\sup_d \mathbb{E}\|w\|^{2p} < \infty$ for every p. On Ω_F, the within-view and cross-view covariances of every block type have operator norm at most C, and for every F-measurable w independent of the block, $\mathbb{E}[|x_{ij}^\top w|^p \mid F, w] \le C_p\|w\|^p$ and $\mathbb{E}[\|e_i\|^p \mid F, w] \le C_p(1 + \|w\|^p)$ for every p.*

Proof. $\frac{\lambda}{2}\|w\|^2 \le L(w) \le L(0) = \frac{1}{2N}\|y\|^2 = \frac{1}{2n}\sum_i t_i^2$, and the moments of the empirical mean of $t_i^2 \sim \chi_1^2$ are bounded. On Ω_F, $\|G_F\| \le 9(1 + \sqrt{\beta})^2$, $\|D_F\| \le \|\sigma'\|_\infty$, $\|A_k\| \le \|\sigma\|_\infty$, and $\|\mathscr{R}\| \le \|Q_F^{-1}\|\,\|G_F\| \le 2\|G_F\|$. The Hadamard sums are bounded because $\sum_k \|A_k\|^2 \le \|\sigma\|_\infty^2$ and, for positive semidefinite A, B, $\|A \circ B\| \le \max_r A_{rr}\,\|B\|$, so that $\|\mathscr{R}^{\circ k}\| \le \|\mathscr{R}\|$ for every k (a correlation matrix has unit diagonal). For the actual block, $x_{ij}^\top w$ is a function of the Gaussian vector (a_i, z_{ij}) with Lipschitz constant $\|\sigma'\|_\infty\|Fw\| \le C\|w\|$ and mean zero (oddness), so its moments are bounded by Gaussian concentration [9, Theorem 5.6]; for the Gaussian block types the same bound is the Gaussian moment bound with the covariance estimate. $e_i = t_i \mathbf{1}_2 - X_i w$ is handled coordinatewise. □

Lemma 6.6 (leverage fluctuations)**.** *Conditional on the leave-block data (which fix F, B and w_-), on Ω_F, for the actual block and for the # block,*

$$\mathbb{E}\big[\|Q_i - \bar{Q}_i\|_F^2 \,\big|\, \mathcal{F}_{-i}\big] \le \frac{C}{d}, \qquad \bar{Q}_i = \mathbb{E}[Q_i \mid \mathcal{F}_{-i}],$$

and $\bar{Q}_i^A = \bar{Q}_i^\#$. The same holds for $R_i = \frac{1}{N} X_i B S_0 B X_i^\top$ with any fixed symmetric S_0 with $\|S_0\| \le C$, and $\|(I + Q)^{-1} - (I + \bar{Q})^{-1}\| \le \|Q - \bar{Q}\|$ for positive semidefinite $Q, \bar{Q}$.

Proof. $(Q_i)_{ab} = \frac{1}{N} x_{ia}^\top B x_{ib}$. For the actual block it is a function of the Gaussian vector $(a_i, z_{i1}, z_{i2}) \in \mathbb{R}^{3d}$ whose gradient with respect to a_i is $\frac{1}{N}\big(F\,\mathrm{diag}(\sigma_a')Bx_{ib} + F\,\mathrm{diag}(\sigma_b')Bx_{ia}\big)$ (and similarly for z_{i1}, z_{i2}), of norm at most $\frac{C}{N}\|F\|\lambda^{-1}\sqrt{m}\,\|\sigma\|_\infty \le C/\sqrt{d}$; the Gaussian Poincaré inequality [9, Theorem 3.20] gives the variance bound. For the # block, $(Q_i)_{ab}$ is a quadratic form in a Gaussian vector; its gradient has norm at most $\frac{C}{N}\|F\|\lambda^{-1}(\|x_{ia}^{\#}\| + \|x_{ib}^{\#}\|)$, whose second moment is $O(m/N^2) = O(1/d)$ by Lemma 6.5, and the Poincaré inequality applies again. $\bar{Q}_i^A = \bar{Q}_i^{\#}$ because $\mathbb{E}[x_{ia}^\top B x_{ib} \mid \mathcal{F}_{-i}] = \mathrm{Tr}(B\,\mathrm{Cov}(x_{ia}, x_{ib} \mid F))$ and the covariances match by Proposition 6.4. The argument for R_i is identical with BS_0B in place of B. The Lipschitz bound for the inverse is the resolvent identity $(I+Q)^{-1} - (I+\bar{Q})^{-1} = (I+Q)^{-1}(\bar{Q}-Q)(I+\bar{Q})^{-1}$ with both resolvents of norm at most one. □

Lemma 6.7 (from the exact Gaussian block to the surrogate)**.** *Let $C_{ab}^{\#}$, C_{ab}^{G} $(a,b \in \{1,2\})$ be the within- and cross-view covariances of the # and G blocks conditional on F, and $c^{\#} = D_F F^\top \xi$, $c^G = \mu_e F^\top \xi$ their teacher cross-covariances. Then*

$$\mathbb{E}\Big[\mathbf{1}_{\Omega_F} \max_{a,b} \|C_{ab}^{\#} - C_{ab}^{G}\|_{\mathrm{op}}\Big] + \mathbb{E}\Big[\mathbf{1}_{\Omega_F}\|c^{\#} - c^{G}\|\Big] \le C\sqrt{\frac{\log d}{d}},$$

and the same bound holds for the clean covariances and cross-covariances of $x^{0,\#}$ and $x^{0,G}$.

Proof. By (21) and (16), $C_{11}^{\#} - C_{11}^{G} = s^2(D_F G_F D_F - \kappa(1)^2 G_F) + (\Sigma_{\mathrm{com}} + \Sigma_{\mathrm{priv}} - \eta(1) I)$ and $C_{12}^{\#} - C_{12}^{G} = (D_F G_F D_F - \kappa(1)^2 G_F) + (\Sigma_{\mathrm{com}} - \gamma(1) I)$. Write $\Sigma_{\mathrm{com}} + \Sigma_{\mathrm{priv}} = \sum_{k\ge 3 \text{ odd}} A_k \mathscr{R}^{\circ k} A_k = \mathrm{diag}(\eta(q_r)) + \Theta$ and $\Sigma_{\mathrm{com}} = \mathrm{diag}(\gamma(q_r)) + \Theta_\rho$, where Θ, Θ_ρ are the off-diagonal parts: the diagonal of $\mathscr{R}^{\circ k}$ is one, so the diagonal entries are $\sum_{k\ge 3} a_k(q_r)^2 = \eta(q_r)$ and $\sum_{k\ge 3} a_k(q_r)^2 \rho^k = \gamma(q_r)$. Three estimates: (i) $\|D_F - \kappa(1) I\| = \max_r |\kappa(q_r) - \kappa(1)| \le C \max_r |q_r - 1|$ on Ω_F by Lemma 6.3, and likewise $\|\mathrm{diag}(\eta(q_r)) - \eta(1) I\|$ and $\|\mathrm{diag}(\gamma(q_r)) - \gamma(1) I\|$; since $q_r \sim \chi^2_d/d$ are m sub-exponential variables, $\mathbb{E}\max_r |q_r - 1| \le C\sqrt{\log m/d}$ [56, Section 2.8]. (ii) $D_F G_F D_F - \kappa(1)^2 G_F = (D_F - \kappa(1) I) G_F D_F + \kappa(1) G_F (D_F - \kappa(1) I)$ has operator norm at most $C \max_r |q_r - 1|$ on Ω_F. (iii) For $r \ne s$, $|\Theta_{rs}| \le \sum_{k\ge 3} |a_k(q_r) a_k(q_s)| |\mathscr{R}_{rs}|^k \le |\mathscr{R}_{rs}|^3 \sqrt{V(q_r) V(q_s)} \le \|\sigma\|_\infty^2 |\mathscr{R}_{rs}|^3$, and $\mathscr{R}_{rs} = \langle f_r, f_s\rangle / (\|f_r\| \|f_s\|)$ is the cosine of the angle between two independent Gaussian vectors in $\mathbb{R}^d$, so $\mathbb{E}\mathscr{R}_{rs}^6 \le C/d^3$. Hence $\mathbb{E}\|\Theta\|_{\mathrm{op}} \le \mathbb{E}\|\Theta\|_F \le \big(\sum_{r\ne s} \mathbb{E}\Theta_{rs}^2\big)^{1/2} \le C m\, d^{-3/2} \le C d^{-1/2}$, and the same for Θ_ρ. The teacher term is $(D_F - \kappa(1) I) F^\top \xi$ with $\|F^\top \xi\| \le \|F\|$. The clean case is identical with $\Delta = 0$. □

6.4 Proof of Theorem 6.1

Proof of (17). For $0 \le i \le n$ let $L^{(i)}$ be the objective in which blocks $1, \dots, i$ are of type G and blocks $i+1, \dots, n$ are actual, so $L^{(0)} = L_A$ and $L^{(n)} = L_G$, and let $L^{(i,\#)}$ be the objective in which block i is of type #, blocks $< i$ of type G and blocks $> i$ actual. Then

$$\mathbb{E}\varphi(L_A^\star) - \mathbb{E}\varphi(L_G^\star) = \sum_{i=1}^{n} \Big(\mathbb{E}\varphi(L^{(i-1)\star}) - \mathbb{E}\varphi(L^{(i,\#)\star})\Big) + \sum_{i=1}^{n} \Big(\mathbb{E}\varphi(L^{(i,\#)\star}) - \mathbb{E}\varphi(L^{(i)\star})\Big).$$

Fix i and let $\mathcal{F}_{-i}$ be the σ-field of F and of all blocks other than i; the three objectives compared share the leave-block minimum $L_-^\star$, the matrix B and the vector w_-. By Lemma 6.2, $L^\star = L_-^\star + \delta$ with $\delta = \frac{1}{2N} e_i^\top (I + Q_i)^{-1} e_i$ computed from the block of the relevant type, and a Taylor expansion gives

$$\big|\mathbb{E}[\varphi(L_-^\star + \delta) - \varphi(L_-^\star + \delta') \mid \mathcal{F}_{-i}]\big| \le \|\varphi'\|_\infty \big|\mathbb{E}[\delta - \delta' \mid \mathcal{F}_{-i}]\big| + \|\varphi''\|_\infty\, \mathbb{E}[\delta^2 + \delta'^2 \mid \mathcal{F}_{-i}].$$

By Lemma 6.5, $\mathbb{E}[\delta^2 \mid \mathcal{F}_{-i}] \le \frac{1}{4N^2}\mathbb{E}[\|e_i\|^4 \mid \mathcal{F}_{-i}] \le \frac{C(1+\|w_-\|^4)}{N^2}$ on Ω_F, so the second-order terms sum to $C\|\varphi''\|_\infty/d$ after taking expectations. For the first-order terms, write $Z = (t_i, u_1, u_2)$ with $u_a = x_{ia}^\top w_-$, so $e_i = t_i \mathbf{1}_2 - (u_1, u_2)^\top$, and freeze the leverage: $\delta = \frac{1}{2N}\psi_{\bar{Q}}(Z) + \frac{1}{2N} e_i^\top \big((I+Q_i)^{-1} - (I+\bar{Q}_i)^{-1}\big) e_i$ with $\psi_{\bar{Q}}(Z) = e_i^\top (I+\bar{Q}_i)^{-1} e_i$ a quadratic form in Z.

Actual to #. The conditional second moments of Z under the actual and the # block coincide (Proposition 6.4, with w_- being $\mathcal{F}_{-i}$-measurable), and $\bar{Q}_i^A = \bar{Q}_i^\#$ (Lemma 6.6); hence $\mathbb{E}[\psi_{\bar{Q}}(Z^A) \mid \mathcal{F}_{-i}] = \mathbb{E}[\psi_{\bar{Q}}(Z^\#) \mid \mathcal{F}_{-i}]$ exactly. The correction term is bounded by $\frac{1}{2N}\|e_i\|^2\|Q_i - \bar{Q}_i\|$, whose conditional expectation is at most $\frac{1}{2N}\big(\mathbb{E}\|e_i\|^4\big)^{1/2}\big(\mathbb{E}\|Q_i - \bar{Q}_i\|_F^2\big)^{1/2} \le \frac{C(1+\|w_-\|^2)}{N\sqrt{d}}$ on Ω_F, by Lemmas 6.5 and 6.6, for either block type. Summed over the n steps this is $C\|\varphi'\|_\infty d^{-1/2}$.

to G. Both blocks are Gaussian conditional on $\mathcal{F}_{-i}$, and $\mathbb{E}[\psi_{\bar{Q}}(Z) \mid \mathcal{F}_{-i}]$ is a fixed linear combination (with coefficients bounded by one) of $\mathbb{E}t_i^2 = 1$, $\mathbb{E}[t_i u_a] = w_-^\top c_a$ and $\mathbb{E}[u_a u_b] = w_-^\top C_{ab} w_-$, while $(\bar{Q}_i)_{ab} = \frac{1}{N}\operatorname{Tr}(BC_{ab})$. By Lemma 6.7, on Ω_F the moments differ by at most $C(1+\|w_-\|^2)\max_{ab}\|C_{ab}^\# - C_{ab}^G\| + C\|w_-\|\|c^\# - c^G\|$, and the frozen leverages by at most $\frac{1}{N}\|B\|_{\mathrm{tr}}\max_{ab}\|C_{ab}^\# - C_{ab}^G\| \le \frac{m}{N\lambda}\max_{ab}\|C_{ab}^\# - C_{ab}^G\|$, which enters ψ through the Lipschitz bound of Lemma 6.6 multiplied by $\|e_i\|^2$. Taking expectations and using Lemma 6.5 for the moments of w_- and e_i, each of the n steps contributes at most $\frac{C}{N}\sqrt{\log d/d}\,\|\varphi'\|_\infty$ to the first-order sum.

Off Ω_F the two expectations differ by at most $2\|\varphi\|_\infty \Pr(\Omega_F^c) \le C\|\varphi\|_\infty e^{-cd}$. Adding the three contributions gives (17). □

Proof of (18). Let $M = \lceil d \rceil$ and let $(t_l^0, a_l^0)_{l \le M}$ be an independent clean block, with actual features $x_l^{0,A} = \sigma(F^\top a_l^0)$; put $c_0 = \frac{1}{M}\sum_l (t_l^0)^2$, $r_0 = \frac{1}{M}\sum_l t_l^0 x_l^{0,A}$, $S_0 = \frac{1}{M}\sum_l x_l^{0,A}(x_l^{0,A})^\top$ and $\widehat{R}(w) = c_0 - 2r_0^\top w + w^\top S_0 w$, the empirical clean risk of w on the block with actual features. Four comparisons, each in L^1 after multiplication by $\|\varphi'\|_\infty$ (with the second-order and off-Ω_F terms as before):

(1) $R_A(w_A^\star \mid F)$ against $\widehat{R}(w_A^\star)$: conditional on F and $w_A^\star$ the block is independent of $w_A^\star$, and $\widehat{R}(w) - R_A(w \mid F)$ is an average of M independent centred variables with variance at most $C(1+\|w\|^4)$ by Lemma 6.5; so the L^1 difference is at most $CM^{-1/2} \le Cd^{-1/2}$.

(2) $\widehat{R}(w_A^\star)$ against $\widehat{R}(w_G^\star)$: the Lindeberg scheme of the previous proof with the statistic $\widehat{R}(w^\star)$. Conditional on $\mathcal{F}_{-i}$ and the clean block, with $v_- = B(S_0 w_- - r_0)$ and $R_i = \frac{1}{N}X_i B S_0 B X_i^\top$, the second formula in (19) gives the exact increment

$$\widehat{R}(w^\star) - \widehat{R}(w_-) = \frac{1}{N}\Big[2e_i^\top (I+Q_i)^{-1} X_i v_- + e_i^\top (I+Q_i)^{-1} R_i (I+Q_i)^{-1} e_i\Big],$$

a quadratic form in $Z' = (t_i, u_1, u_2, v_1, v_2)$ with $v_a = x_{ia}^\top v_-$, once Q_i and R_i are frozen at their conditional means. On the event $\{\|S_0\| \le C\}$ (of probability $1 - Ce^{-cd}$ by [56, Theorem 4.6.1] and Lemma 6.5), $\|v_-\| \le C(1+\|w_-\|)$, the second moments of Z' match exactly between the actual and the # block (Proposition 6.4, applied to the $\mathcal{F}_{-i}$-measurable vectors w_- and v_-), the frozen means of Q_i and R_i match exactly, the fluctuations of Q_i and R_i are $O(1/d)$ in mean square (Lemma 6.6), and the #-to-G step is controlled by Lemma 6.7 exactly as before, with $\frac{1}{N}\|BS_0B\|_{\mathrm{tr}} \le \frac{m\|S_0\|}{N\lambda^2}$. Summing over n blocks gives $C\|\varphi'\|_\infty\sqrt{\log d/d} + C\|\varphi''\|_\infty/d$.

(3) $\widehat{R}(w_G^\star)$ against $R_\#(w_G^\star \mid F) = \mathbb{E}[(t^0 - (x^{0,A})^\top w_G^\star)^2 \mid F, w_G^\star]$: the empirical-to-population step (1) again, $O(d^{-1/2})$.

(4) $R_\#(w_G^\star \mid F)$ against $R_G(w_G^\star \mid F)$: by Proposition 6.4 the first equals the population risk under the exact clean Gaussian row, and the two Gaussian clean risks differ by $|2(w_G^\star)^\top(c_0^\# - c_0^G)| + |(w_G^\star)^\top(C_0^\# - C_0^G)w_G^\star|$, which is $O(\sqrt{\log d/d})$ in L^1 by Lemma 6.7 and Lemma 6.5.

Adding the four bounds gives (18). □

Remark 6.8. The factor $\sqrt{\log d}$ in (17)–(18) comes from a single place, the maximum of the m column norms $|q_r - 1|$ in Lemma 6.7(i). Replacing that maximum by an average over the coordinates of w_- requires a delocalisation bound $\mathbb{E}|w_{-,r}|^4 \le Cd^{-2}$ on the coordinates of the leave-block minimiser, under which the rate becomes $d^{-1/2}$ [45]; the equivalence itself does not depend on it. Two of the objects in the proof are measured in Section 10: the leverage matrix Q_i of Lemma 6.2, whose fluctuation Lemma 6.6 bounds by C/d, and the Hermite coefficients (15), which determine the surrogate and whose residual variances $\eta(1) - \gamma(1)$ and $\gamma(1)$ are the quantities behind Proposition 10.1 there.

Part II
Finite-size measurements

7 Target 3: the finite-block certificate

7.1 What the proof uses

For zero signal the excess risk of the minimum-norm interpolant is a quadratic form in the noise, and its expectation over the noise is the leverage functional

$$\frac{V(X)}{\sigma^2} \;=\; \mathbb{E}_U\big\|k(U,X)K^{-1}\big\|^2 \;=\; \mathrm{Tr}\big(K^{-1}GK^{-1}\big), \qquad G=\sum_s \lambda_s^2 N_s P_s, \tag{22}$$

the second equality because $\mathbb{E}_U[k(X,U)k(U,X)] = \sum_s \lambda_s^2 N_s P_s$ by Mercer and the addition theorem. Writing $K_s = \lambda_s N_s P_s$ and $t_s = \mathrm{Tr}(K^{-1}K_s)$, Cauchy–Schwarz in the Frobenius inner product gives $t_s^2 \le nN_s\lambda_s q_s$ with $q_s = \mathrm{Tr}(K^{-1}K_sK^{-1})$, and summing over a window B of degrees,

$$\frac{V(X)}{\sigma^2} \;\ge\; \frac{T_B^2}{n\,D_B}, \qquad T_B=\sum_{s\in B} t_s, \quad D_B=\sum_{s\in B} N_s. \tag{23}$$

The proof of Target 3 [45] takes B to be the *cumulative* block $0,\dots,L$, writes $D_L = \sum_{s\le L} N_{d,s}$, $\delta_L = D_L/n$, $T_L = \mathrm{Tr}(K^{-1}\sum_{s\le L} K_s)$, and bounds the leverage the block captures in expectation over the design,

$$\mathbb{E}\,T_L \;\ge\; \frac{n\,D_L}{D_L + n - 1 + \tau_{>L}/\lambda_{d,L}}, \qquad \tau_{>L}=\sum_{s>L}\tau_s, \tag{24}$$

with no concentration step: only operator monotonicity of the inverse, Cauchy–Schwarz, the Mercer decomposition, the addition theorem and Jensen's inequality enter. In the critical regime the tail ratio $\theta_L = \tau_{>L}/(n\lambda_{d,L})$ is $o(1)$, so (24) reads $\mathbb{E}\,T_L/n \ge \delta_L/(1+\delta_L+o(1))$, and taking expectations in (23) and applying Jensen once more,

$$\mathbb{E}\,\frac{V(X)}{\sigma^2} \;\ge\; \frac{(\mathbb{E}\,T_L/n)^2}{\delta_L} \;\ge\; \frac{\delta_L}{(1+\delta_L)^2} - o(1). \tag{25}$$

For the largest L with $D_L \le n$ the ratio δ_L lies in $(1/5, 1]$ for all large d, by the exact multiplicity ratio and Stirling's formula, and $y \mapsto y/(1+y)^2$ is increasing on $[0,1]$, so

$$\liminf_{d\to\infty} \mathbb{E}\,\frac{V(X)}{\sigma^2} \;\ge\; \frac{1/5}{(6/5)^2} = \frac{5}{36} = c_1,$$

for every fixed $h > 0$. This is the one-window theorem. Splitting the degrees into nested cumulative windows $B_0 \subset \cdots \subset B_J$ with $y_j = D_{B_j}/n$ and $A_j = T_{B_j}/n$, the pathwise reduction (23) applied shell by shell gives $V/\sigma^2 \geq A_0^2/y_0 + \sum_{j\geq 1}(A_j - A_{j-1})^2/(y_j - y_{j-1})$, a convex quadratic form in the A_j, and the same expectation bounds inserted into it, with the window offsets chosen against the phase of $n = 2^d$ relative to the harmonic multiplicities, give the ten-window certificate

$$c_{10} \;=\; \frac{3882594969529}{12500000000000} \;=\; 0.31060759756232\ldots,$$

an exact rational number, again for every fixed $h > 0$ [45]. The one-window constant is Theorem 4.1; the ten-window certificate is a strengthening obtained in [45] by running the argument of Section 4 over ten blocks at once (Remark 4.5), and is quoted here as a comparison value. No random-matrix universality enters at any point. Both constants are independent of h; the measured functional is not, and the distance between them is measured below.

7.2 What is measured

Every object in the last subsection is computable from the Gram matrix alone. P_s evaluated at the matrix of inner products follows the dimension-d Legendre recurrence $(s+d-2)P_{s+1}(t) = (2s+d-2)tP_s(t) - sP_{s-1}(t)$, so no spherical harmonic is ever formed; the projection kernel $\Pi_L = \sum_{s\leq L} N_s P_s$ gives $\|S_L - I\|_F^2 = \|\Pi_L\|_F^2/n^2 - D_L$ for the block's orthonormal feature matrix Φ_L and $S_L = \frac{1}{n}\Phi_L^\top \Phi_L$; and G in (22) is another accumulation over the same stream. For each design and each L the driver records D_L, δ_L, T_L/n, the certificate $(1/\delta_L)(T_L/n)^2$, and the ratio of the measured $\|S_L - I\|_F^2$ to its exact expectation

$$\mathbb{E}\big\|S_L - I\big\|_F^2 = \frac{D_L(D_L-1)}{n}, \tag{26}$$

an elementary identity that serves as a check on the Gram-only implementation. It reports $V(X)/\sigma^2$ twice: once as a Monte-Carlo average over fresh test points, once exactly through (22).

Carrying both is what exposes a numerical trap. At $h = 1/2$ and $h = 1$ the two derivations agree to a fraction of a percent. At $h = 2$ the form $\mathrm{Tr}(K^{-1}GK^{-1})$ evaluated as written does not agree with the Monte-Carlo value, and the disagreement grows with d; the identity is not at fault, its evaluation is. In the eigenbasis of K the trace reads $\sum_k \tilde{G}_{kk}/w_k^2$, so rounding of size $\varepsilon\|G\|$ in the entries of G is amplified by $\varepsilon\,\kappa(K)^2$, not by $\varepsilon\kappa$: the inverse appears twice. The Gaussian kernel at $h = 2$ is the smoothest of the three bandwidths and K the worst conditioned, and Table 1 shows $\varepsilon\kappa^2$ climbing from 3 at $d = 9$ to 2×10^3 at $d = 11$, where the trace form has departed from the Monte-Carlo control by two percent while the factored evaluation described next stays within a tenth of a percent of it. Factoring G alone does not help, because the rounding directions of G carry eigenvalues of order $\varepsilon\|G\|$ that are still divided by w_k^2. The evaluation that survives uses the *same* factors for both matrices: with $N_sP_s(T) = Y_sY_s^\top$ from one eigendecomposition per degree, $K = \sum_s \lambda_s Y_s Y_s^\top$ and $G = \sum_s \lambda_s^2 Y_s Y_s^\top$, so that

$$V/\sigma^2 = \sum_{s\leq L} \lambda_s^2 \,\|K^{-1}Y_s\|_F^2,$$

a form with one inverse in which rounding enters squared, and in which each factor's own rounding directions are separated from its true spectrum by many orders of magnitude and dropped. The exact column of Table 3 is evaluated this way wherever $\varepsilon\kappa^2$ exceeds 10^{-4}, and both forms are kept in the records with the criterion. The point is general: an exact trace identity carrying two

Table 1: The two evaluations of (22) at $h = 2$, $n = 2^d$, means over the designs of Table 3. $\varepsilon\kappa^2$ is machine precision times the squared condition number of K; the trace form is $\mathrm{Tr}(K^{-1}GK^{-1})$ as written, the factor form uses the shared per-degree factors, and the Monte-Carlo column averages $\|k(U, X)K^{-1}\|^2$ over 4,000 fresh points per design.

d	designs	$\varepsilon\kappa^2$	trace form	factor form	Monte Carlo
9	40	3.2	3.1941	3.1941	3.1925
10	24	84	3.8754	3.8753	3.8757
11	16	2.0×10^3	2.5691	2.5157	2.5138

Table 2: The cumulative blocks at $d = 10$, $n = 2^{10}$, $h = 1$, means over 36 designs (the standard deviation across designs of T_L/n and of the certificate is at most 0.0012; that of the identity ratio falls from 0.18 at $L = 1$ to 0.004 at $L = 6$). $\delta_L = D_L/n$; the certificate is $(1/\delta_L)(T_L/n)^2$; the two columns headed by functions of δ_L are the theorem's asymptotic lower bounds (24) and (25) for T_L/n and for the certificate; the identity column is the measured $\|S_L - I\|_F^2$ divided by the exact (26); the last column marks the block the one-window theorem uses, the largest L with $D_L \le n$.

L	D_L	δ_L	T_L/n	$\frac{\delta_L}{1+\delta_L}$	certificate	$\frac{\delta_L}{(1+\delta_L)^2}$	(26)	theorem's block
1	11	0.0107	0.0107	0.0106	0.0107	0.0105	0.982	
2	65	0.0635	0.0634	0.0597	0.0633	0.0561	1.003	
3	275	0.2686	0.2636	0.2117	0.2588	0.1669	1.001	
4	935	0.9131	0.7330	0.4773	0.5884	0.2495	1.000	yes
5	2717	2.6533	0.9639	0.7263	0.3502	0.1988	1.000	
6	7007	6.8428	0.9964	0.8725	0.1451	0.1112	0.999	

inverses is not an exact number once $\varepsilon\kappa^2$ is of order one, and for a smooth kernel that happens at modest n.

Four things are visible in Tables 2 and 3. The identity (26) holds: the ratio sits within 2% of one from $D_L = 65$ on and within a tenth of a percent from $D_L = 275$ on. The leverage the block captures exceeds the expectation bound (24) at every L with $D_L \le n$, and by a wide margin at the theorem's block: at $d = 10$ the block $L = 4$ has $\delta_L = 0.913$ and captures $T_L/n = 0.733$ against the bound's 0.477, so at these sizes (24) is the loose step. The certificate at the theorem's block sits above $c_1 = 5/36$ at every dimension and bandwidth measured, between 0.21 and 0.76; its variation with d is a sawtooth that follows δ_{L_-}, which is set by where 2^d falls between two block boundaries, and not a trend. And the quantity being bounded exceeds the ten-window constant c_{10} by a factor between 1.7 and 2.1 at $h = 1/2$, between 5.5 and 6.1 at $h = 1$, and between 8.1 and 12.5 at $h = 2$: the functional is monotone in the bandwidth while the constant is not, and the gap between them is where an h-dependent constant would live.

The two derivations of V/σ^2 agree throughout: the largest relative discrepancy over all designs is 1.2×10^{-2}, and the mean values agree to within 0.13 percent. The Mercer mass $\sum_s \tau_s$ recovers to 1.000000 at $h \ge 1$ and to between 0.9980 and 0.9994 at $h = 1/2$, where the eigenvalue tail reaches past the harmonic truncation.

Table 3: The bounded functional $V(X)/\sigma^2$ at $n = 2^d$, mean $\pm$ standard deviation over independent designs, and its ratio to the ten-window constant $c_{10} = 0.3106$. The last three columns give the one-window theorem's block L_-, the largest L with $D_L \le n$, its δ_L, and the mean certificate $(1/\delta_L)(T_L/n)^2$ at that block, to be read against $c_1 = 5/36 = 0.1389$. The cells $(d, h) = (12, 2)$, $(13, 2)$ and all of $d = 14$ were not run.

d	h	$V(X)/\sigma^2$	designs	$/\, c_{10}$	L_-	δ_{L_-}	certificate at L_-
9	0.5	0.5371 ± 0.0067	40	1.73	3	0.410	0.214
9	1	1.7743 ± 0.0468	60	5.71	3	0.410	0.373
9	2	3.1925 ± 0.1125	40	10.28	3	0.410	0.403
10	0.5	0.5621 ± 0.0032	24	1.81	4	0.913	0.285
10	1	1.8357 ± 0.0246	36	5.91	4	0.913	0.588
10	2	3.8757 ± 0.0775	24	12.48	4	0.913	0.760
11	0.5	0.5891 ± 0.0023	16	1.90	4	0.628	0.279
11	1	1.7087 ± 0.0144	24	5.50	4	0.628	0.517
11	2	2.5138 ± 0.0269	16	8.09	4	0.628	0.598
12	0.5	0.6125 ± 0.0016	8	1.97	4	0.422	0.250
12	1	1.7047 ± 0.0096	12	5.49	4	0.422	0.390
13	0.5	0.6384 ± 0.0010	4	2.06	5	0.978	0.320
13	1	1.8797 ± 0.0061	6	6.05	5	0.978	0.611

8 The dimension gap between Targets 2 and 3

Targets 2 and 3 are the same construction at two ends of one number. Let $k(d, n) = \min\{s : N_{d,s} \ge n\}$ be the critical degree and

$$\delta \;=\; \frac{\log\left(N_{d,k}/n\right)}{\log d}.$$

Target 2's sequence has $\log N_{d,k}/\log d \to R + \delta_\star$ with $\delta_\star = 1/\log 2 - 1 = 0.442695\ldots$, so it runs at $\delta \to \delta_\star > 0$, and the proof's tail-isotropy step turns on exactly that positive gap: the harmonic dimension above the cutoff outruns the sample by $d^{\delta_\star}$, which is what makes $\|K_{\mathrm{T}}/\tau - I\| \to 0$ for the tail block K_{T} of the kernel matrix and its Mercer mass τ. Target 3's $n = 2^d$ puts the crossing at $k \sim c_\star d$ with $c_\star = 0.2938\ldots$ solving $(1 + c)\log(1 + c) - c\log c = \log 2$, and there the gap closes. The same coordinate organises the large-dimensional theory of kernel ridge regression [22, 41, 36]: with $n \asymp d^\gamma$ the fractional part of γ is the gap of the last resolved harmonic block.

The natural experiment is to fix d, sweep n, and watch the noise risk against δ. One feature of the geometry has to be respected first: δ is not monotone in n. Each time n crosses a multiplicity $N_{d,k}$ the critical degree advances and δ resets to the top of its range, so a plain sweep in n traces a sawtooth in δ while the risk climbs steadily with sample size for reasons that have nothing to do with the gap. Comparing across that sawtooth measures n, not δ. The comparison has to be made *within* a block, at fixed critical degree.

Done that way the ordering is clean. Table 4 shows one dimension in full: at $d = 8$, $h = 1/2$, at fixed critical degree k, the mean noise risk falls monotonically as δ rises, in every one of the six blocks, over four independent designs per cell. Table 6 summarises the same sweep at seven dimensions from $d = 8$ to $d = 64$, all at $h = 1/2$: over 20 blocks with at least three values of the gap, 8 of the 211 ordered pairs of cells are inverted, and every inversion lies either in a block with $N_{d,k} \le 1287$, where the noise risk is at most 0.39 and the two inverted cells differ by at most 0.012, or at $d = 64$, where only two designs were run per cell. Across blocks the risk rises with n, as it

Table 4: The dimension gap at fixed critical degree, $d = 8$, $h = 1/2$, four designs per cell. Each row is one harmonic block: n runs over the sample sizes whose critical degree is k, and the entries are the mean noise risk $\mathcal{R}_{\text{noise}}/\sigma^2$ at each value of the gap δ. Within every row the risk falls as δ rises.

k	$N_{d,k}$	$\delta = 0.02$	0.05	0.10	0.20	0.30
5	672	0.772	0.754	0.705	0.697	0.627
6	1386	1.125	1.088	0.989	0.885	0.819
7	2640	1.482	1.337	1.286	1.207	1.126
8	4719	1.760	1.668	1.609	1.487	
9	8008	2.054	1.983	1.967	1.802	
10	13013	2.428	2.389	2.257	2.147	

Table 5: The tail isotropy $\|K_{\text{T}}/\tau - I\|$ in the cells of Table 4.

k	$N_{d,k}$	$\delta = 0.02$	0.05	0.10	0.20	0.30
5	672	1.897	1.893	1.844	1.533	1.447
6	1386	2.130	2.158	1.913	1.765	1.615
7	2640	2.302	2.285	2.214	1.828	1.650
8	4719	2.439	2.421	2.174	2.023	
9	8008	2.648	2.435	2.310	1.976	
10	13013	2.783	2.664	2.388	2.105	

must. From $d = 12$ on the ladder reaches $\delta = 0.44$, the value of $\delta_\star$, and at $d = 24$, 32 and 48 it reaches $\delta = 0.6$ (Table 7); the risk keeps falling through both, with no turnover at any dimension.

The tail isotropy $\|K_{\text{T}}/\tau - I\|$, the quantity Target 2's proof drives to zero, follows the same ordering (Table 5), and strictly so in all but one of the blocks of Table 6. It is an operator norm, a maximum over eigenvalues, and a maximum is a noisier estimate than an average at the same number of designs; the one block in which it inverts does so between the two smallest gaps, where its means differ by one percent.

What this does and does not show is worth stating carefully. The two theorems constrain the two ends of the order parameter — one proves a vanishing risk along a sequence with δ bounded away from zero, the other a positive floor along a sequence where the gap closes — but neither implies anything about the monotonicity of a finite-size curve at fixed d. The measured ordering is consistent with them and is the shape one would expect if δ really is the controlling variable; it is evidence for that reading, not a consequence of either proof. All of it is at one bandwidth.

Table 6: Within-block ordering at every dimension measured, $h = 1/2$. A block is a critical degree k at which at least three values of δ were run; a pair of cells in a block is inverted when the cell with the larger gap has the larger mean noise risk. The isotropy column counts the blocks in which $\|K_{\mathrm{T}}/\tau - I\|$ falls strictly along the whole row; the last column names the blocks that carry an inverted risk pair.

d	blocks	designs per cell	inverted pairs	isotropy monotone	blocks with an inversion
8	6	4	0 / 48	5 / 6	–
12	4	4	2 / 43	4 / 4	$k = 3$, $k = 4$
16	3	4	0 / 35	3 / 3	–
24	2	4	0 / 27	2 / 2	–
32	2	4	2 / 27	2 / 2	$k = 2$
48	2	4	0 / 16	2 / 2	–
64	1	2	4 / 15	1 / 1	$k = 2$

Table 7: The blocks whose ladder reaches $\delta \geq 0.44$: mean noise risk $\mathcal{R}_{\mathrm{noise}}/\sigma^2$ at $h = 1/2$, four designs per cell, out to $\delta = 0.6$ where it was run.

d	k	$N_{d,k}$	$\delta = 0.02$	0.05	0.10	0.20	0.30	0.44	0.60
12	4	1287	0.393	0.388	0.392	0.369	0.318	0.271	–
12	5	4004	0.619	0.576	0.568	0.517	0.487	0.424	–
16	4	3740	0.347	0.331	0.316	0.300	0.277	0.228	–
16	5	14688	–	0.478	0.468	0.436	0.407	0.366	–
24	3	2576	0.160	0.153	0.150	0.135	0.117	0.096	0.079
24	4	17250	–	–	0.247	0.230	0.212	0.188	–
32	3	5952	0.135	0.135	0.120	0.117	0.103	0.084	0.070
48	3	19552	–	–	–	0.095	0.087	0.075	0.067
64	2	2079	0.042	0.045	0.045	0.035	0.025	0.025	–

9 Target 4: the Veronese reduction and its two lemmas

9.1 The reduction

Write $x = ru$ with u uniform on S^{d-1} and $r^2 \sim \chi^2_d$ independent, and identify $z(x)$ with $Q = (xx^\top - I)/\sqrt{2} \in \mathrm{Sym}_d$ in an orthonormal Frobenius basis whose first element is $E_0 = I/\sqrt{d}$. Then the trace coordinate is $c(r) = (r^2 - d)/\sqrt{2d}$ and each traceless coordinate is $a(r)h_j(u)$ with $a(r) = r^2/\sqrt{d(d+2)}$ and $h_j(u) = \sqrt{d(d+2)/2}\; u^\top E_j u$, so that after an orthogonal change of feature coordinates

$$ZO = D\,[\,b,\; H\,], \qquad D = \mathrm{diag}(a_i), \quad b = D^{-1}c, \tag{27}$$

with b *exactly* independent of H, and every row of H centred, isotropic and of fixed norm $m = p-1$. The scalar column satisfies $b = \sqrt{(d+2)/2}\,(r^2 - d)/r^2$, whence for $d > 4$

$$\mathbb{E}b = -\frac{\sqrt{2(d+2)}}{d-2}, \qquad \mathrm{Var}(b) = \frac{d^2(d+2)}{(d-4)(d-2)^2} = 1 + \frac{10}{d} + O(d^{-2}),$$

and $\max_i |a_i - 1| = O_p(\sqrt{\log d/d})$, so the hard edge of Z and of $[b, H]$ coincide in the limit. In the basis of an SVD of H the smallest eigenvalue of $[b, H][b, H]^\top$ obeys an exact rank-one secular equation, and the leave-one-out normal has trace weight $\alpha^2 = (1 + \|H_-^{-1}b_-\|^2)^{-1}$ [45].

The reduction names two lemmas whose truth would materially advance the transfer:

Table 8: Target 4 lemma diagnostics. A is the unit-Frobenius-norm feature-space normal of $p-1$ rows; u_1, u_2, u_3 are the bottom left singular vectors of H and $\mathbf{1} \in \mathbb{R}^p$. Entries are mean $\pm$ standard deviation over the designs, with the 90th percentile of the overlap beside it; extremes are not reported, since the width of a min–max grows with the number of designs. The seventh column is $s_{\min}(Z)/s_{\min}([b, H])$, the effect of the diagonal scaling D in (27). A third column tending to a positive limit is (L-b); a fifth column that does not grow like $\sqrt{d}$ is (L-a).

d	p	$d\,\mathrm{Tr}(A^4)$	$\sqrt{d}\,\|A\|_{\mathrm{op}}$	$\lvert u_j^\top \mathbf{1}\rvert$	p90	$s_{\min}$ ratio	designs
20	210	2.37 ± 0.24	2.03 ± 0.15	0.71 ± 0.54	1.47	0.815 ± 0.040	1200
30	465	2.24 ± 0.15	2.02 ± 0.12	0.73 ± 0.56	1.49	0.873 ± 0.022	900
40	820	2.18 ± 0.09	2.02 ± 0.10	0.74 ± 0.56	1.54	0.902 ± 0.014	600
60	1830	2.12 ± 0.06	2.02 ± 0.08	0.77 ± 0.59	1.59	0.934 ± 0.007	420
80	3240	2.09 ± 0.04	2.01 ± 0.06	0.79 ± 0.60	1.66	0.951 ± 0.005	270
100	5050	2.07 ± 0.03	2.02 ± 0.05	0.79 ± 0.60	1.63	0.961 ± 0.004	240
120	7260	2.06 ± 0.03	2.02 ± 0.05	0.80 ± 0.58	1.53	0.967 ± 0.002	36

(L-a) $|u_j^\top \mathbf{1}| = o_p(\sqrt{d})$ for the bottom left singular vectors u_j of H (this suffices to kill the mean shift of b; $O_p(1)$ is plausible);

(L-b) $\mathbb{E}\,\mathrm{Tr}(A^4) = O(1/d)$ for the feature-space normal A of $p-1$ rows, normalized to $\|A\|_F = 1$.

9.2 What is measured

The identity (27) is verifiable exactly, and is verified: $s_{\min}(Z)$ and $s_{\min}(D[b, H])$ agree, in the median over designs, to 1.0×10^{-11} relative at $d = 20$ and to 6.3×10^{-9} at $d = 120$, the discrepancy growing with d as the conditioning does; every row of H has $\|h(u)\|^2 = m$ to 10^{-15}, and the mean and variance of the scalar column b match their closed forms to within 1.4 percent at every dimension. The asymptotic half — that dropping D does not move the edge — converges slowly but visibly: the ratio $s_{\min}(Z)/s_{\min}([b, H])$ in Table 8 rises from 0.815 at $d = 20$ to 0.967 at $d = 120$, with a spread across designs that shrinks from 0.04 to 0.002, and d times its distance from one stays between 3.7 and 3.9 over the whole range, so the diagonal scaling moves the edge by about $3.9/d$ at these sizes.

Both lemmas look true, and (L-a) in its stronger form (Table 8). The overlaps $|u_j^\top \mathbf{1}|$ show no growth with d at all — 0.71 at $d = 20$ and 0.80 at $d = 120$, with a standard deviation near 0.6 throughout — which is $O_p(1)$, not merely the $o_p(\sqrt{d})$ the reduction asks for.

(L-b) is sharper than a glance suggests. $d\,\mathrm{Tr}(A^4)$ is not constant (Table 9): the values at $d = 20$ and $d = 40$ are 24 combined standard errors apart, so a flat fit is rejected outright. It is converging, with a $1/d$ correction. Fitting $d\,\mathrm{Tr}(A^4) = a + b/d$ to the two end points alone gives

$$d\,\mathrm{Tr}(A^4) \;=\; 1.990 + \frac{7.66}{d},$$

which reproduces the middle point, never used in the fit, to within 0.30 of its standard error; the weighted three-point fit reads $a = 1.990 \pm 0.010$ and $b = 7.65 \pm 0.32$ with a chi-squared of 0.06 on one degree of freedom. So $\mathrm{Tr}(A^4) = \Theta(1/d)$, which is (L-b), with the constant identified rather than merely bounded.

The constant repays a second look, because 2 is not an arbitrary number here. Were the $p-1$ rows generic rather than Veronese, their normal would be a uniform direction on the Frobenius sphere of Sym_d; the spectrum of such a direction is semicircular [57, 2], and $d\,\mathrm{Tr}(A^4)$ then tends

Table 9: Lemma (L-b) at the three dimensions with the largest design counts. The null is the exact generic-direction value (28); the excess is the measured mean minus the null, with the standard error of the mean and the excess in units of it; the last column is d times the excess.

d	designs	$d\,\mathrm{Tr}(A^4)$	null (28)	excess	in s.e.	$d\times$excess
20	1200	2.3728 ± 0.0069	2.0328	0.340 ± 0.007	49	6.80
30	900	2.2436 ± 0.0049	2.0256	0.218 ± 0.005	45	6.54
40	600	2.1813 ± 0.0039	2.0207	0.161 ± 0.004	41	6.43

to the fourth moment of the semicircle law, the Catalan number $C_2 = 2$. At finite d the null is exact: a uniform direction is $U = G/\|G\|_F$ with G a GOE matrix, whose norm is independent of its direction, so

$$d\,\mathbb{E}\,\mathrm{Tr}(U^4) = \frac{d\,\mathbb{E}\,\mathrm{Tr}(G^4)}{\mathbb{E}\|G\|_F^4} = \frac{d\,(2d^3+5d^2+5d)}{4p(p+2)}, \qquad p = \frac{d(d+1)}{2}, \tag{28}$$

using $\mathbb{E}\,\mathrm{Tr}(G^4) = 2d^3 + 5d^2 + 5d$ for the GOE normalised to unit off-diagonal variance, by Wick's formula, and $\|G\|_F^2 \sim 2\chi_p^2$. The measured intercept 1.990 ± 0.010 sits 1.0 standard errors from 2. To leading order the leave-one-out normal behaves like a generic direction, and the limit in (L-b) is the semicircle's.

What is not generic is the approach to it. At each dimension the measured value exceeds the null (28) by forty to fifty standard errors (Table 9), so the gap is not a sampling artefact, and multiplying the excess by d gives values that drift gently downward rather than sitting still: a limit with its own $1/d$ correction, the two-point fit $6.05 + 14.9/d$ landing the middle value within 0.07 of a standard error. So the excess over a uniform direction is itself $\Theta(1/d)$, and

$$\mathrm{Tr}(A^4) \;=\; \frac{2}{d} + \frac{c}{d^2} + \cdots, \qquad c \approx 6,$$

where a uniform direction contributes under one to that second coefficient. The Veronese structure of the rows is invisible in the leading term of $\mathrm{Tr}(A^4)$ and enters only at the next one. For (L-b), which asks only for the order, that is a wide margin; for anyone setting out to prove it, the leading calculation is one about a generic direction in Sym_d, and the rows' structure can be carried as a correction.

9.3 Why the constant is two

The value the constant took is not a numerical accident, and following it gives a statement about A that is stronger than (L-b), simpler than (L-b), and — unlike (L-b) — of a shape that a proof can attack directly. Two exact facts do the work, and both are elementary.

Proposition 9.1 (Conjugation invariance)**.** *The law of A is invariant under $A \mapsto OAO^\top$ for every $O \in O(d)$. Consequently $A = U\Lambda U^\top$ with U Haar on $O(d)$ independent of Λ, and every question about A is a question about its eigenvalue law alone.*

Proof. For $v \in \mathbb{R}^p$ let $M_v \in \mathrm{Sym}_d$ be its un-vectorization, so that $z(x)^\top v = \langle M_v, xx^\top - I\rangle/\sqrt{2}$. Then $z(Ox)^\top v = \langle M_v, Oxx^\top O^\top - I\rangle/\sqrt{2} = \langle O^\top M_v O, xx^\top - I\rangle/\sqrt{2}$, so replacing every input x_i by Ox_i maps the row space of Z through the linear map $M \mapsto OMO^\top$, which is orthogonal for the Frobenius inner product. The orthogonal complement of the row space maps the same way, and the inputs Ox_i have the law of the x_i. □

Proposition 9.2 (The Veronese feature vector is isotropic)**.** $\mathbb{E}\, z(x)z(x)^\top = I_p$ *exactly, for every* d*.*

Proof. The diagonal coordinates have $\mathbb{E}[(x_a^2-1)^2/2] = (3-1)/2 = 1$ and the off-diagonal ones $\mathbb{E}[x_a^2 x_b^2] = 1$. Every cross term vanishes: $\mathbb{E}[(x_a^2-1)(x_b^2-1)] = 0$ for $a \neq b$ by independence, and any term carrying an odd power of some coordinate vanishes by symmetry. □

Both scalars in Proposition 9.2 are forced, and it is worth seeing why, because it says what would have to fail for the leading order to stop being generic. The group $O(d)$ acts on Sym_d preserving the splitting $\mathrm{Sym}_d = \mathbb{R}I \oplus \mathrm{Sym}_d^0$ into the scalars and the traceless part, and for $d \geq 3$ these are inequivalent irreducibles. Since $z(Ox) = Oz(x)O^\top$ and Ox has the law of x, the operator $\mathbb{E}\, zz^\top$ commutes with the action, so by Schur's lemma it is scalar on each summand:

$$\mathbb{E}\, zz^\top = \alpha\, P_{\mathbb{R}I} + \beta\, P_{\mathrm{Sym}_d^0}.$$

No entry-by-entry computation can produce anything else, and isotropy is the single equality $\alpha = \beta$. For the Gaussian row both are 1: along the traceless block a pure off-diagonal coordinate is $x_a x_b$ with variance 1, and along $I/\sqrt{d}$ the row reads $(|x|^2 - d)/\sqrt{2d}$, whose variance is 1 because χ_d^2 has variance $2d$. So $\alpha = \beta$ is a property of the Gaussian and not of the Veronese map, and it is the only thing about the row law that the leading order uses. The same argument applied to the normal itself gives its second moment: $\mathbb{E}\, A \otimes A$ is $a_d P_{\mathbb{R}I} + b_d P_{\mathrm{Sym}_d^0}$ with $a_d = O(1/d)$ and $b_d = (1-a_d)/(p-1)$, so that $\mathbb{E}\langle A, B\rangle^2 = (1 + O(1/d))/(p-1)$ for every fixed unit traceless B [45]; this is the second-moment shadow of the statement below, at the scale a Haar-distributed direction would give.

Proposition 9.2 is what makes the leading order generic. Had the $p-1$ rows been isotropic *Gaussian* vectors in $\mathbb{R}^p$, their normal would have been exactly uniform on the unit Frobenius sphere of Sym_d; a uniform direction is a Frobenius-normalized GOE matrix, whose spectrum is semicircular, and the semicircle's moments are the Catalan numbers. The rows are not Gaussian — they are quadratic in x — but they are exactly isotropic, and that is the part the leading order sees.

Proposition 9.1 also says what cannot be measured. Every $\mathrm{Tr}(A^{2k})$, the operator norm, the Kolmogorov–Smirnov distance to the semicircle — every statistic of A alone that is invariant under conjugation — is a function of the eigenvalues, and the eigenvectors carry no information beyond being Haar. So no such statistic can distinguish the Veronese normal from any other conjugation-invariant matrix with the same spectrum, and whatever the design leaves in A beyond its spectrum can only be seen in a statistic that involves A *and* the data: the alignment of an eigenvector with one of the x_i, or $x^\top A x$ at a fresh point. That is where a proof of the conjecture may not go, and where a sharper description of A must.

So the statement the measurement is pointing at is this.

Conjecture 9.3. $\sqrt{d}\, A$ converges in moments to the standard semicircle law on $[-2, 2]$; equivalently, for every $k \geq 1$,

$$d^{\,k-1}\, \mathbb{E}\, \mathrm{Tr}(A^{2k}) \;\longrightarrow\; C_k = \frac{1}{k+1}\binom{2k}{k},$$

the odd moments vanishing by symmetry. Lemma (L-b) is the case $k = 2$, where $C_2 = 2$.

Table 10 is the test proper — the same normal taken to the sixth, eighth and tenth moments and out to dimensions the full hard-edge driver cannot reach, the null vector being a linear solve rather than a decomposition. Every moment sits above its Catalan number and falls toward it as d

Table 10: Conjecture 9.3 under test. A is the same unit-Frobenius normal as in Table 8; each column reports $d^{k-1}\operatorname{Tr}(A^{2k})$ against the Catalan number it should converge to, and the last column the edge against 2. Entries are mean ± standard error over independent designs. The null vector is obtained by a linear solve and checked by its residual $\|Zv\|$, which is below 10^{-10} on every design reported here.

d	designs	$d\ \operatorname{Tr} A^4$ ($C_2 = 2$)	$d^2\ \operatorname{Tr} A^6$ ($C_3 = 5$)	$d^3\ \operatorname{Tr} A^8$ ($C_4 = 14$)	$d^4\ \operatorname{Tr} A^{10}$ ($C_5 = 42$)	$\sqrt{d}\,\|A\|_{\mathrm{op}}$ (2)
20	320	2.371 ± 0.012	7.458 ± 0.098	27.005 ± 0.621	106.621 ± 3.671	2.034 ± 0.008
30	270	2.258 ± 0.008	6.678 ± 0.062	22.665 ± 0.364	83.736 ± 1.995	2.036 ± 0.007
40	200	2.187 ± 0.006	6.198 ± 0.047	20.082 ± 0.272	70.724 ± 1.485	2.029 ± 0.007
60	140	2.118 ± 0.005	5.752 ± 0.036	17.784 ± 0.204	59.649 ± 1.087	2.021 ± 0.007
80	60	2.091 ± 0.005	5.565 ± 0.029	16.743 ± 0.149	54.266 ± 0.709	2.007 ± 0.006
100	40	2.071 ± 0.004	5.440 ± 0.030	16.151 ± 0.166	51.722 ± 0.850	2.009 ± 0.010
140	10	2.055 ± 0.009	5.332 ± 0.056	15.575 ± 0.269	48.934 ± 1.212	2.019 ± 0.013
160	6	2.054 ± 0.008	5.319 ± 0.053	15.500 ± 0.265	48.577 ± 1.206	2.009 ± 0.017
200	6	2.037 ± 0.005	5.227 ± 0.029	15.100 ± 0.132	46.912 ± 0.572	2.013 ± 0.006

grows, and the edge $\sqrt{d}\,\|A\|_{\mathrm{op}}$ sits above 2 throughout, between 2.007 and 2.036 over $20 \le d \le 200$, with no growth. The pipeline has a control (Table 11): replacing the Veronese rows by i.i.d. Gaussian or Rademacher rows in $\mathbb{R}^p$ makes the null direction exactly uniform on the sphere, so its un-vectorization is a Frobenius-normalized GOE matrix and its fourth moment must be the exact null (28); it is, to within 2.3 standard errors at every dimension run, while the Veronese law sits far above at the same dimensions. The whole of the Veronese excess is a property of the feature law rather than of the measurement.

The limit is an extrapolation, and it is worth being exact about what the data support. Table 12 reads off the intercept two ways over the whole range of dimensions, weighting each dimension by the inverse squared standard error of its mean. A two-term fit $a + b/d$ sits below the Catalan number at every k, by under one standard error at $k = 2$ and by between two and four at $k = 4$ and $k = 5$, with a chi-squared that reaches 9.6 on 7 degrees of freedom at $k = 5$: the higher moments carry a visible $1/d^2$ term that a two-term fit cannot absorb and loads onto the intercept. Adding that term, the three-term fit $a + b/d + e/d^2$ brings the intercept to within 0.2 standard errors of the Catalan number at each of the four moments, and its $1/d$ coefficient at $k = 2$, $b = 7.2 \pm 0.7$, agrees with the value the hard-edge driver measures independently in Table 9. So the reading is the careful one: the data are consistent with the Catalan limits and the limits are approached, not attained — at $d = 200$, the largest reached, $d\ \operatorname{Tr}(A^4)$ is still 2.037, 7 standard errors above 2. Above $d = 100$ the ladder rests on ten, six and six designs.

Moments are a confirmation, and four of them agreeing is still four numbers; a law that is not the semicircle can match them. The observation that comes out differently if the conjecture is false is the shape of the whole spectrum, and the statistic for that is the Kolmogorov–Smirnov distance between the empirical distribution of $\sqrt{d}\,\lambda$ and the semicircle law itself. If the conjecture holds this must fall to zero; if the limit is some other law it must flatten at a positive value, however well the low moments behave. It falls (Table 13), and what makes that readable is the control beside it, since the absolute value of a Kolmogorov–Smirnov distance at d eigenvalues means nothing on its own: Gaussian and Rademacher rows give a null direction that is exactly uniform, hence genuinely semicircular in the limit, and at the two dimensions where both arms were run the Veronese distance is 1.06 to 1.10 times the Gaussian-row distance. A laptop-scale run of the same

Table 11: The pipeline control. The same null-vector computation with the Veronese rows replaced by i.i.d. Gaussian and by i.i.d. Rademacher rows in $\mathbb{R}^p$, whose normal is exactly a uniform direction; its $d\ \mathrm{Tr}(A^4)$ must equal (28), which is 2.0329, 2.0258, 2.0207 and 2.0141 at $d = 20, 30, 40, 60$. Mean $\pm$ standard error over the designs.

d	rows	designs	$d\ \mathrm{Tr}\,A^4$	$d^2\ \mathrm{Tr}\,A^6$	$d^3\ \mathrm{Tr}\,A^8$	$d^4\ \mathrm{Tr}\,A^{10}$	$\sqrt{d}\,\|A\|_{\mathrm{op}}$
20	Veronese	320	2.371 ± 0.012	7.458 ± 0.098	27.005 ± 0.621	106.621 ± 3.671	2.034 ± 0.008
20	Gaussian rows	200	2.041 ± 0.011	5.258 ± 0.066	15.311 ± 0.334	48.263 ± 1.588	1.861 ± 0.008
20	Rademacher rows	200	2.037 ± 0.010	5.263 ± 0.065	15.443 ± 0.333	49.207 ± 1.600	1.867 ± 0.008
30	Veronese	270	2.258 ± 0.008	6.678 ± 0.062	22.665 ± 0.364	83.736 ± 1.995	2.036 ± 0.007
30	Gaussian rows	150	2.039 ± 0.008	5.263 ± 0.050	15.384 ± 0.254	48.667 ± 1.208	1.916 ± 0.008
30	Rademacher rows	150	2.025 ± 0.007	5.186 ± 0.045	15.023 ± 0.229	47.080 ± 1.106	1.905 ± 0.008
40	Veronese	200	2.187 ± 0.006	6.198 ± 0.047	20.082 ± 0.272	70.724 ± 1.485	2.029 ± 0.007
40	Gaussian rows	100	2.018 ± 0.006	5.118 ± 0.040	14.616 ± 0.208	44.984 ± 0.995	1.916 ± 0.008
40	Rademacher rows	100	2.002 ± 0.008	5.046 ± 0.051	14.373 ± 0.259	44.323 ± 1.229	1.917 ± 0.009
60	Veronese	140	2.118 ± 0.005	5.752 ± 0.036	17.784 ± 0.204	59.649 ± 1.087	2.021 ± 0.007
60	Gaussian rows	60	2.022 ± 0.006	5.144 ± 0.039	14.725 ± 0.184	45.305 ± 0.812	1.944 ± 0.007
60	Rademacher rows	60	2.013 ± 0.006	5.077 ± 0.034	14.384 ± 0.157	43.792 ± 0.700	1.938 ± 0.007

Table 12: The intercept of the moment ladder of Table 10, computed from the records over every dimension measured, each weighted by the inverse squared standard error of its mean. For each k: a two-term fit $a + b/d$, its intercept's distance from the Catalan number in standard errors and its chi-squared over degrees of freedom; then the same for a three-term fit $a + b/d + e/d^2$, with its $1/d$ coefficient.

k	C_k	two-term a	s.e. from C_k	χ^2/dof	three-term a	b	s.e. from C_k	χ^2/dof
2	2	1.997 ± 0.004	-0.7	4.4/7	2.000 ± 0.006	7.2 ± 0.7	+0.0	4.1/6
3	5	4.959 ± 0.024	-1.7	4.0/7	4.995 ± 0.040	44.8 ± 4.7	-0.1	2.8/6
4	14	13.669 ± 0.121	-2.7	5.8/7	13.970 ± 0.201	213.9 ± 25.3	-0.2	2.3/6
5	42	39.904 ± 0.581	-3.6	9.6/7	41.956 ± 0.952	917.7 ± 127.7	-0.0	2.2/6

construction at $d = 20$, with 200 designs per arm, gives 0.0919 ± 0.0012 against 0.0848 ± 0.0012, the same ratio. The Veronese spectrum sits a tenth further from the semicircle than an exactly uniform direction does, at every dimension where the comparison exists, and the two fall together. A law converging to something else would have separated from that control rather than tracking it.

Nor is the conjecture special to the Gaussian row. Take the inputs uniform on the unit sphere instead and centre inside the traceless block, $z(x) = xx^\top - I/d$; then the design never leaves Sym_d^0, the scalar block of Proposition 9.2 is absent, isotropy is automatic by Schur, and A is the unit traceless symmetric matrix whose quadratic form kills the design, $x_i^\top A x_i = 0$ for $i \le \dim \mathrm{Sym}_d^0 - 1$. In laptop-scale runs of that construction the four moments converge to the same Catalan numbers, and the corrections are the same size: with 7,373, 2,650 and 925 designs at $d = 24, 32, 40$ the excess over C_k is 0.93 to 1.03 times the excess of the Gaussian-input normal at the same d, read off the $1/d + 1/d^2$ model fitted to the records, for every k from 2 to 5; a direct pairing of the two constructions at the same d with a few hundred designs each gives ratios between 0.9 and 1.2 with standard errors of five to ten percent. Same sign, same $1/d$ shape, the same size to within the errors. Two row laws, two ambient spaces, one limit.

Where the non-Gaussianity sits is visible, and it has an exact formula. Write U for a unit-Frobenius symmetric matrix and $u_1, \dots, u_d$ for its eigenvalues. Then $\langle z, U \rangle = \sum_i u_i (g_i^2 - 1)/\sqrt{2}$ with g standard Gaussian, a weighted sum of independent centred χ_1^2 variables; since $(g^2 - 1)$ has

Table 13: Kolmogorov–Smirnov distance between the empirical law of $\sqrt{d}\,\lambda$ and the semicircle law on $[-2, 2]$, mean $\pm$ standard error over the designs whose records carry the statistic, by row law.

d	rows	designs	KS distance
40	Veronese	100	0.0515 ± 0.0009
40	Gaussian rows	100	0.0484 ± 0.0009
40	Rademacher rows	100	0.0481 ± 0.0008
60	Veronese	100	0.0379 ± 0.0006
60	Gaussian rows	60	0.0346 ± 0.0007
60	Rademacher rows	60	0.0359 ± 0.0007
80	Veronese	30	0.0300 ± 0.0009
100	Veronese	20	0.0247 ± 0.0009
200	Veronese	6	0.0141 ± 0.0005

variance 2 and fourth cumulant 48,

$$\operatorname{kurt}\langle z, U\rangle \;=\; 3 + 12\,\operatorname{Tr}(U^4). \tag{29}$$

The two coordinate values are the extremes of (29): a pure diagonal coordinate has $\operatorname{Tr}(U^4) = 1$ and kurtosis 15, a pure off-diagonal coordinate has $\operatorname{Tr}(U^4) = \frac{1}{2}$ and kurtosis 9. Along a *typical* direction $\operatorname{Tr}(U^4)$ is the fourth moment of a Frobenius-normalized GOE spectrum, (28) divided by d, so the kurtosis along a uniformly random direction is $3 + 24/d$ to leading order; a random direction mixes $\Theta(d^2)$ coordinates and the mixture is asymptotically Gaussian. Along the null direction of a design the row is slightly more leptokurtic than along an independent direction: for the sphere construction, where the typical value falls just below three, the pairs read 2.84 against 3.21 at $d = 8$, 2.89 against 3.23 at $d = 10$, 2.94 against 3.18 at $d = 14$ and 2.97 against 3.11 at $d = 20$, the gap closing with d, while the two variances agree to three digits at every d — a direct check that the null direction is isotropic at second order (laptop, 12,753 to 97,056 designs).

What the fourth moment does *not* explain is the size of the correction, and the sphere construction is the reason. Its row $xx^\top - I/d$ has no $|x|^2$ fluctuation, and its kurtosis along a typical direction falls just *below* three, against $3 + 24/d$ for the Gaussian row. Two row laws whose kurtoses straddle the Gaussian value, one leptokurtic by a factor of nearly two at $d = 10$ and the other slightly platykurtic, produce corrections agreeing to within a tenth. A fourth moment cannot do that. Matching directly on the kurtosis makes the point sharper still. Take rows with *independent* coordinates of variance one and kurtosis κ — the three-point law $\Pr(\pm\kappa^{1/2}) = 1/(2\kappa)$, $\Pr(0) = 1 - 1/\kappa$ realises every $\kappa \geq 1$ — and the excess over C_k is decided by κ and nothing else: at $d = 14$, over about 10^5 designs per arm, a three-point law at $\kappa = 3$ and a Gaussian agree to within half a standard error at $k = 2, 3, 4, 5$, while moving κ from 1 to 9 multiplies the excess by 2.3; at $d = 24$ the same comparison holds for three matched pairs (uniform against three-point at $\kappa = 1.8$, Gaussian against three-point at $\kappa = 3$, Laplace against three-point at $\kappa = 6$, all within two standard errors over 17,752 to 71,636 designs per arm) and the same move in κ multiplies the excess by only 1.15, so the κ-dependence is itself a higher-order term in $1/d$ than the excess. The Veronese row at $d = 14$ has kurtosis $3 + 24/14 \approx 4.7$ along a typical direction and an excess of 0.46 at $k = 2$ against 0.041 for independent coordinates matched to that kurtosis, an order of magnitude larger. With independent coordinates the correction is a fourth-moment functional; the Veronese row does not have independent coordinates, it is rank one, and that is the structure the two constructions share and the fourth moment does not see. A row $(XX^\top - rI)/\sqrt{2r}$ with X a $d \times r$ Gaussian matrix is a sum of r independent rank-one forms, has kurtosis $3 + 12\,\operatorname{Tr}(U^4)/r$

along U, and its excess falls roughly as $1/r$ toward the generic value: 0.46, 0.23, 0.13, 0.085 at $r = 1, 2, 4, 8$ ($k = 2$, $d = 14$). Within one dependence structure the kurtosis tunes the correction; across structures at a fixed kurtosis it does not, and the Veronese row's $r = 1$ is the extreme. A derivation of the constant b has to come from the rank-one dependence, not from a kurtosis. All of the numbers in this paragraph are laptop-scale computations at $d \le 24$, recorded with their scripts beside the run records.

Remark 9.4 ((L-b) follows from a delocalization bound, and the loss is a factor two)**.** Since $\mathrm{Tr}(A^2) = 1$,

$$\mathrm{Tr}(A^4) = \sum_i \lambda_i^4 \;\le\; \Big(\max_i \lambda_i^2\Big) \sum_i \lambda_i^2 = \|A\|_{\mathrm{op}}^2,$$

so (L-b) is implied by $\mathbb{E}\|A\|_{\mathrm{op}}^2 = O(1/d)$ — a statement about one eigenvalue rather than about the whole spectrum, and the kind of delocalization estimate the random-matrix literature is built to supply. It is worth knowing how much that costs, because a crude bound that loses an unbounded factor is not a route. It loses exactly two. Under Conjecture 9.3 the edge is $2/\sqrt{d}$ and the fourth moment $2/d$, so the bound gives $4/d$ against a truth of $2/d$; measured, the ratio of $\|A\|_{\mathrm{op}}^2$ to $\mathrm{Tr}(A^4)$ climbs from 1.75 at $d = 20$ to 1.99 at $d = 200$, converging to the two that the semicircle predicts. So an operator-norm route to (L-b) is not lossy in order and is lossy by a factor of two in the constant, which the reader attacking it should know before choosing between it and the moment route.

Remark 9.5 (The Gram matrix as a sum of independent rank-one terms)**.** The Gram matrix of the reduction, $\sum_i z(X_i)z(X_i)^\top$, is a sum of independent rank-one matrices with heavily dependent entries within each summand, which is the setting of the matrix concentration inequalities of Brailovskaya and van Handel [10] and of the Gaussian theory they rest on [5], and outside that of edge universality for matrices with independent entries. In that framework the spectrum is controlled by the free model with the same mean and covariance — here the Marchenko–Pastur law at aspect ratio one [38], whose hard edge is the one the conjecture names — provided the norm of a single summand is small against the variance parameter of the sum. The variance parameter is exact: by the same Schur decomposition, $\mathbb{E}[\|z\|^2 zz^\top]$ has the two eigenvalues $(d^2 + 13d + 16)/2$ on $\mathbb{R}I$ and $(d^2 + 9d + 16)/2$ on Sym_d^0, so for $X = \sum_{i \le p-1}(z_i z_i^\top - I)$ one has $\sigma(X)^2 = \|\mathbb{E}X^2\| = (p-1)(d^2 + 13d + 14)/2$. The summand norm is $\|z(x)\|^2 = (|x|^4 - 2|x|^2 + d)/2$ with $|x|^2 \sim \chi_d^2$, of mean p and with $|x|^4/2$ as its leading term, and in a laptop-scale sample at $d = 10$ to 44 its 99th percentile runs from 3.1 down to 2.1 times $\sigma(X)$ while its maximum stays between 4.1 and 12 times $\sigma(X)$: the boundedness parameter is of the order of the variance parameter at every reachable d, not small against it. Its tail is exactly what makes the inverse smallest singular value fail to be integrable at fixed d (the remark after Table 14); a truncation of that tail is the natural device, and the rate at which the truncated summand becomes negligible is the quantitative question the framework poses.

9.4 Does the reduced matrix look like a Wishart at the edge?

The reduction is exact and the two lemmas look true, so the question that remains is whether $[b, H]$ actually obeys the target law. That is directly testable: Table 14 compares the empirical distribution of each statistic against $1 - e^{-\sqrt{t}-t/2}$ by a Kolmogorov–Smirnov distance, with the 95% critical value $1.36/\sqrt{N}$ at N designs beside it. After the reduction the matrix is statistically indistinguishable from the square real Wishart hard edge at every dimension from $d = 20$ on, while Z itself is rejected at $d = 20$, 30 and 40 and becomes consistent only from $d = 60$, where

Table 14: Kolmogorov–Smirnov distance of the empirical law of $p\,s_{\min}^2$ to the square real Wishart hard edge $1-e^{-\sqrt{t}-t/2}$, for Z, for the reduced matrix $[b,H]$, and for the $p\times(p-1)$ block H alone (tested against the same law, to which it is not expected to conform; see the text). The critical value is $1.36/\sqrt{N}$ at the 95% level; the verdict is for Z and for $[b,H]$.

d	designs	Z	$[b,H]$	H	critical	verdict (Z / $[b,H]$)
20	1200	0.104	0.024	0.397	0.039	rejected / consistent
30	900	0.078	0.024	0.401	0.045	rejected / consistent
40	600	0.058	0.020	0.399	0.056	rejected / consistent
60	420	0.054	0.035	0.431	0.066	consistent / consistent
80	270	0.058	0.037	0.421	0.083	consistent / consistent
100	240	0.035	0.047	0.427	0.088	consistent / consistent
120	36	0.114	0.099	0.379	0.227	consistent / consistent

the critical value has also grown. The test has finite resolution and it should be quoted with it: at 1,200 designs a Kolmogorov–Smirnov test sees a discrepancy only above about 0.04 in the cumulative distribution, so this is evidence against distortions larger than four percentage points at $d=20$, and against progressively larger ones at the higher dimensions with fewer designs. The two statistics differ by exactly the diagonal scaling D, whose effect the theory shows vanishes and which at $d=20$ still contributes a factor $s_{\min}(Z)/s_{\min}([b,H])=0.815$; squaring that moves the whole distribution of $p\,s_{\min}(Z)^2$ left by a third, which is what the first column registers. The finite-size discrepancy therefore sits in the one place the proof already controls.

Remark 9.6. The same test applied to H alone rejects at a distance near 0.4 at every dimension, and that number should not be read as evidence about anything. H is $p\times(p-1)$, one column short of square, and the hard-edge law depends on the aspect deficit through the Bessel parameter $\alpha=|\text{rows}-\text{cols}|$ [53]; $1-e^{-\sqrt{t}-t/2}$ is the $\alpha=0$ law. Testing an $\alpha=1$ matrix against it rejects by construction. The column is recorded only to say that it is uninformative.

Remark 9.7. For every fixed $d\geq 2$, $\mathbb{E}\,s_{\min}(Z)^{-1}=\infty$ [45]. The rows span Sym_d almost surely, so $\det Z\neq 0$ almost surely, and the least singular value of a square matrix with i.i.d. rows has a positive linear lower-tail coefficient at the origin in every fixed dimension [26], which forces the inverse first moment to diverge. This is not evidence against the conjecture — the Gaussian square ensemble diverges the same way [19], and the conjectured law $1-e^{-\sqrt{t}-t/2}$ has exactly the linear lower tail in $s_{\min}$ that causes it — but it forbids any argument that asks for uniform integrability of the full inverse smallest singular value, as opposed to a truncated one or one with finitely many bottom modes removed.

It is visible in the records, and it is a warning about reporting. Table 15 splits the designs at each dimension into disjoint blocks of 150 and computes $(\sqrt{p}\,s_{\min}(Z))^{-1}$ in each. A sample mean of 150 independent draws of a quantity with finite variance varies by a few percent from block to block; these block means vary by factors of 2 to 7, with coefficients of variation between 0.35 and 0.84, because each is estimating the smallest draw in its block rather than any population quantity, while the block medians spread by 7 to 26 percent (coefficient of variation 0.04 to 0.07). That is what a divergent first moment looks like from inside a finite sample, and it is why the tables in this section report medians and standard deviations rather than means of inverse quantities.

Remark 9.8. The lower tail is where the conjecture lives, so it is worth looking at directly rather than through a single Kolmogorov–Smirnov number. Table 16 takes the 2nd, 5th, 10th and 20th

Table 15: Block means and block medians of $(\sqrt{p}\, s_{\min}(Z))^{-1}$ over disjoint blocks of 150 designs, with the coefficient of variation across blocks.

d	blocks	block means	c.v.	block medians	c.v.
20	8	5.5–33.0	0.82	2.12–2.66	0.07
30	6	5.8–38.0	0.80	1.92–2.24	0.06
40	4	5.3–10.2	0.35	1.95–2.15	0.04
60	2	3.7–14.7	0.84	1.94–2.07	0.05

Table 16: The lower tail of $p\, s_{\min}([b, H])^2$: the mass assigned by $1 - e^{-\sqrt{t}-t/2}$ below the empirical 2nd, 5th, 10th and 20th percentiles, at every dimension with at least 200 designs.

d	designs	at p2	at p5	at p10	at p20
20	1200	0.025	0.058	0.102	0.205
30	900	0.022	0.043	0.088	0.181
40	600	0.021	0.044	0.111	0.201
60	420	0.025	0.067	0.102	0.194
80	270	0.021	0.051	0.088	0.187
100	240	0.025	0.042	0.127	0.227

percentiles of $p\, s_{\min}([b, H])^2$ and reports the mass the conjectured law assigns below each of them; the nominal values are 0.02, 0.05, 0.10, 0.20. The law is a little heavy in the extreme tail at $d = 20$ and a little light at $d = 30$, with no monotone drift in d — a pattern consistent with sampling noise at these design counts rather than with a systematic distortion, and consistent with the distances of Table 14.

Remark 9.9. A measurement cannot prove either lemma, and neither can it prove the hard edge. What it can do is tell the next attempt where the difficulty is likely to be. On this evidence neither lemma is the obstruction, and the reduced matrix already passes the test the conjecture is about at the smallest dimension measured; what is missing is a hard-edge theorem for the harmonic block, not a reason to doubt the target.

10 Target 1: the surrogate's residuals, and why no Stein comparison is needed

10.1 A Hermite identification

The proof of Theorem 6.1 matches a Gaussian surrogate whose same-view variance and cross-view covariance agree with the activation's, with the shared and private residual coefficients of Dhifallah and Lu [17]. The identification of those coefficients is a Hermite expansion, and their nonnegativity is immediate from it.

Proposition 10.1. *Let $s = \sqrt{1+\Delta^2}$ and $\rho = 1/(1+\Delta^2)$, and expand $\varphi(\alpha) = \sigma(s\alpha) = \sum_{k\ge 0} c_k \mathrm{He}_k(\alpha)/\sqrt{k!}$ for $\alpha \sim N(0,1)$. Write $V = \mathbb{E}\varphi^2$ and $C_{12} = \mathbb{E}[\varphi(A_1)\varphi(A_2)]$ for a standard bivariate normal pair of correlation ρ. Then the shared and private residual variances of the matched surrogate are*

$$c_{\mathrm{sh}} = C_{12} - c_1^2\rho = \sum_{k\ge 2} c_k^2 \rho^k, \qquad c_{\mathrm{pr}} = V - c_1^2 - c_{\mathrm{sh}} = \sum_{k\ge 2} c_k^2\,(1-\rho^k),$$

Table 17: Target 1 diagnostics, $\sigma = \tanh$, $m = 2d$, $n = 3d$, $\lambda = 10^{-2}$; "unit" rows normalize the columns of F, "raw" rows leave them Gaussian. Entries are mean ± standard deviation over the designs; the two risk columns are the gap between the two models' clean risks at their own minimizers and at the same w. A fourth column constant in d is the $\Theta(1/d)$ leverage variance the proof needs.

d	Δ	columns	$d\,\mathrm{Var}(q_{11})$	$\lvert L^*_{\mathrm{act}} - L^*_{\mathrm{sur}}\rvert / L^*$	$\lvert R_{\mathrm{act}} - R_{\mathrm{sur}}\rvert$	same-w	designs
64	0.25	unit	0.49 ± 0.04	0.07 ± 0.05	1.3×10^{-2}	1.4×10^{-3}	360
64	0.25	raw	0.49 ± 0.04	0.08 ± 0.06	1.4×10^{-2}	3.8×10^{-3}	360
64	0.5	unit	0.37 ± 0.03	0.06 ± 0.05	1.9×10^{-2}	1.8×10^{-3}	360
64	0.5	raw	0.37 ± 0.03	0.06 ± 0.05	2.0×10^{-2}	4.8×10^{-3}	360
64	1	unit	0.29 ± 0.02	0.05 ± 0.04	3.0×10^{-2}	2.6×10^{-3}	360
64	1	raw	0.29 ± 0.02	0.05 ± 0.04	3.0×10^{-2}	6.5×10^{-3}	360
128	0.25	unit	0.49 ± 0.04	0.06 ± 0.04	8.5×10^{-3}	1.3×10^{-3}	300
128	0.25	raw	0.49 ± 0.04	0.05 ± 0.05	7.7×10^{-3}	2.4×10^{-3}	100

and both are nonnegative for every $\rho \in [0, 1]$.

Proof. Orthonormality of the Hermite basis gives $V = \sum_k c_k^2$ and, since a bivariate normal pair of correlation ρ satisfies $\mathbb{E}[\mathrm{He}_j(A_1)\mathrm{He}_k(A_2)] = \delta_{jk} k! \rho^k$, also $C_{12} = \sum_k c_k^2 \rho^k$. Subtracting the $k = 1$ terms gives the two displayed series, and $c_0 = 0$ for odd σ. Each series has nonnegative terms for $0 \le \rho \le 1$. □

Numerically, with $\sigma = \tanh$ and $\Delta = 1/2$ (so $\rho = 0.8$): $c_1 = 0.63166$, $V = 0.43503$, $C_{12} = 0.33574$, $c_{\mathrm{sh}} = 0.016546$, $c_{\mathrm{pr}} = 0.019483$, and the two definitions of each agree to machine precision.

10.2 The leverage matrix

The reason the proof of Theorem 6.1 does not need a three-dimensional correlated Stein comparison is that the exact leave-one-block objective increment depends on six scalars, not three — the teacher, the two logits, and the three entries of the 2×2 leverage matrix $Q_{ab} = N^{-1} x_a^\top H_-^{-1} x_b$ — and Q concentrates. The Gaussian Poincaré inequality [9] gives $\mathrm{Var}(Q_{ab} \mid F, H_-) \le C/d$, after which the block functional is quadratic in the three logits and second-moment matching suffices [45]; this is the step at which the argument departs from the Stein-method route of [28].

That is the step measured here. Table 17 reports $d\,\mathrm{Var}(q_{11})$ over fresh blocks, together with the relative gap between the optimized objectives of the actual and surrogate models, the gap between their clean risks, and the gap between the two models' risks at the same w, at $d = 64$ for three augmentation strengths and at $d = 128$ for one, and for two normalizations of the feature matrix: columns of F normalized to unit length, and raw Gaussian columns, whose $O(\sqrt{\log d / d})$ norm fluctuation is one of the two sources the rate ε_d is made of. The product $d\,\mathrm{Var}(q_{11})$ is of order one and depends on Δ, falling from 0.49 at $\Delta = 1/4$ to 0.29 at $\Delta = 1$; the two column normalizations agree to within a hundredth on it, on the objective gap and on the end-to-end risk gap, and differ only in the same-w risk gap, where the raw columns are 2.4 to 2.8 times larger at $d = 64$ and 1.9 times at $d = 128$; and at $\Delta = 1/4$ the value at $d = 128$ equals the value at $d = 64$. One doubling of d is consistent with a constant and cannot distinguish it from a slow drift, and nothing more is claimed for the behaviour in d.

11 An exact same-sample identity, on two emulation corpora

11.1 The identity

A residual kernel correction fitted on top of a data-dependent mean is a construction used in emulation pipelines for radiative transfer [48, 49], following the kernel-flow emulators of [33] built on the kernel-flow method of [43], and the question of when it helps is a question about the same sample being used twice. Fix a realized training sample S; let m_S and k_S be a mean and a kernel that may depend on S in any way whatsoever, $r_S = G - m_S$ the residual function, X the training inputs, $A = K + n\lambda I$, $R = r_S(X)$, $\alpha = A^{-1}R$ and $q(u) = k_S(X, u)$. With $U \sim \mu$ independent of S, put $Q_S = \mathbb{E}_U q(U)q(U)^\top$ and $B_S = \mathbb{E}_U q(U)r_S(U)^\top$. Then [45]

$$\mathcal{R}_1(S) - \mathcal{R}_0(S) \;=\; -2\,\mathrm{Tr}(\alpha^\top B_S) + \mathrm{Tr}(\alpha^\top Q_S \alpha), \tag{30}$$

exactly, with no independence assumed anywhere; the correction improves the conditional population risk if and only if $2\,\mathrm{Tr}(\alpha^\top B_S) > \mathrm{Tr}(\alpha^\top Q_S \alpha)$. A two-point counterexample in the same source shows that a same-sample residual correction can strictly worsen population risk over a whole range of positive ridge, so the sign in (30) is not automatic. A companion bound $\eta \le \kappa\,\delta_{\mathrm{rms}}/(2\sqrt{\lambda})$ controls the L^2 distance η between the correction fitted against out-of-fold training-row predictions and the one fitted against the deployed predictions, in terms of the root-mean-square difference δ_{rms} between the two sets of training-row residuals.

11.2 On the data

The identity is evaluated on the two corpora of the emulation pipelines [48]: the four radiance components $Y_1, \ldots, Y_4$ of the EMIT imaging spectrometer [25] and the three spectral bands of the OCO-2 instrument [21, 50] — the O_2 A-band and the weak and strong CO_2 bands. In each case a neural mean (width 384, depth 3, 120 epochs) is trained and two residual vectors are formed on the training rows: the *consistent* residuals, against the deployed mean's own predictions, and the *out-of-fold* residuals, against predictions from four folds of the same architecture retrained without the row, which are the labels the emulation pipeline fits its correction to. A Matérn-5/2 correction is fitted to each, with one length scale for both, chosen on the validation split for the correction fitted to the consistent residuals, and both are scored on the test split, which supplies the sample from μ. The kernel stage is fitted on a random subsample of 8,000 of the mean's own training rows: the identity is exact for whatever realized sample it is evaluated on, and the cap is what keeps an exact Gram inside memory. The identity, the margin and every "gain" below refer to the consistent arm; the out-of-fold arm is reported beside it in Table 18 and is the subject of the mismatch bound. Neither Q_S nor B_S is ever formed: with K_{Xt} the train-by-test kernel block, $\mathrm{Tr}(\alpha^\top Q_S \alpha) = \|K_{Xt}^\top \alpha\|_F^2/n_{\mathrm{te}}$ and $\mathrm{Tr}(\alpha^\top B_S) = \langle K_{Xt}^\top \alpha, R_{\mathrm{te}}\rangle/n_{\mathrm{te}}$. Eight ridge values from 10^{-8} to 10^{-1} are swept on every lane; there are five seeds per component, except for the strong CO_2 band, which has one. Two corpora are two corpora: the remaining emulation data sets of the same pipelines were not run, and nothing here generalises beyond EMIT and OCO-2.

The identity holds to machine precision: across the 31 lanes and 248 (lane, ridge) pairs the largest relative discrepancy between the two sides of (30) is 4×10^{-12} (Table 18). That is a check on the arithmetic, and it passes.

The sign is where the content is, and it is decided by the component rather than by the ridge alone (Tables 18 and 19). On Y_1 and Y_2 the margin $2\,\mathrm{Tr}(\alpha^\top B) - \mathrm{Tr}(\alpha^\top Q \alpha)$ is positive at every one of the eight ridges on every seed, and the correction reduces the mean's conditional test risk by 25.5 and 22.1 percent at the best ridge. On Y_3 the margin is positive at every ridge from 10^{-5}

Table 18: The same-sample identity on the EMIT components and the OCO-2 bands. "Gain" is the reduction in conditional test risk of the consistent arm — the correction fitted to the residuals against the deployed mean's own training-row predictions — at its best ridge λ^*, as a percentage of the mean's own risk, mean ± standard deviation over the lanes; "phys." is the same gain in the pipeline's physical relative error; "hurt" counts the (lane, ridge) pairs, out of eight per lane, where the margin $2\,\mathrm{Tr}(\alpha^\top B) - \mathrm{Tr}(\alpha^\top Q\alpha)$ is negative. The two "o.o.f." columns are the out-of-fold arm — the correction fitted to the residuals against out-of-fold predictions, the labels the pipeline fits to — scored on the same test block: its gain at the consistent arm's λ^*, and its best gain over the eight ridges. The last column is the worst relative discrepancy between the two sides of (30).

corpus	lanes	gain %	phys. gain %	hurt	o.o.f. at λ^* %	o.o.f. best %	identity
emit Y1	5	25.5 ± 4.2	25.9 ± 3.0	0 / 40	-45.7 ± 17.6	0.4 ± 0.3	2×10^{-13}
emit Y2	5	22.1 ± 2.1	27.1 ± 5.1	0 / 40	-41.3 ± 20.4	0.0 ± 0.1	1×10^{-13}
emit Y3	5	15.3 ± 3.9	22.2 ± 5.3	6 / 40	-34.7 ± 21.4	0.4 ± 0.4	2×10^{-13}
emit Y4	5	0.5 ± 3.0	0.5 ± 1.5	32 / 40	-2.1 ± 3.1	0.0 ± 1.8	1×10^{-13}
oco2 o2	5	0.3 ± 0.1	0.2 ± 0.1	31 / 40	-5.5 ± 1.6	-0.2 ± 0.2	4×10^{-12}
oco2 sco2	1	1.3 ± 0.0	0.6 ± 0.0	4 / 8	-15.1 ± 0.0	-0.2 ± 0.0	8×10^{-14}
oco2 wco2	5	0.3 ± 0.3	0.2 ± 0.2	28 / 40	-3.4 ± 4.3	0.1 ± 0.1	7×10^{-13}

Table 19: Lanes on which the margin $2\,\mathrm{Tr}(\alpha^\top B) - \mathrm{Tr}(\alpha^\top Q\alpha)$ is positive, by ridge value, out of the lanes in the second column.

corpus	lanes	10^{-8}	10^{-7}	10^{-6}	10^{-5}	10^{-4}	10^{-3}	10^{-2}	10^{-1}
emit Y1	5	5	5	5	5	5	5	5	5
emit Y2	5	5	5	5	5	5	5	5	5
emit Y3	5	2	3	4	5	5	5	5	5
emit Y4	5	1	1	1	1	1	1	1	1
oco2 o2	5	0	0	0	1	1	4	1	2
oco2 sco2	1	0	0	0	0	1	1	1	1
oco2 wco2	5	0	0	0	0	3	3	3	3

up, but it is negative at 10^{-8} on 3 seeds, at 10^{-7} on 2 and at 10^{-6} on 1 (Table 19), 6 of the forty pairs in all, with a crossing that is monotone in λ and interior to the swept grid; Y_3 is also the component the correction helps least when it does help, 15.3 percent. On Y_4 the margin is negative at every ridge on four of the five seeds — 32 of forty pairs — and the correction changes the risk by 0.5 percent at the best ridge, which is to say by nothing. The mean's own test risk on that component is an order of magnitude larger than on the other three, at the same rows, epochs and folds: the network fits Y_4 poorly, and the correction spends its capacity on the part of the residual that does not transfer to fresh points, which is the mechanism of the counterexample on the production pipeline itself. On the OCO-2 bands the correction is close to neutral: gains of 0.3, 1.3 and 0.3 percent on the O_2, strong CO_2 and weak CO_2 bands at the best ridge, with the margin negative at every ridge up to 10^{-6} on every lane, negative at 10^{-5} on 10 of the 11 lanes, and of either sign and within 2.2 percent of $\mathcal{R}_0$ from 10^{-4} up. So the harmful regime that the counterexample describes is real and reachable on real data, and the identity locates it per component and per ridge, computed rather than assumed.

One transfer is worth checking rather than assuming. The identity is a statement about squared L^2 risk, while the emulator pipelines report a physical relative error, and an improvement in the first need not survive into the second — a gain concentrated on high-magnitude components can

wash out entirely once errors are taken relative. It survives here. Over the 15 EMIT lanes on Y_1, Y_2 and Y_3 the squared-risk gain averages 21.0 percent and the gain in the physical metric 25.0 percent, and across all 31 lanes of Table 18 the two gains agree in sign on 31. The physical metric inverts the principal-component and standardisation maps and returns the mean relative L^2 error in physical units, the pipeline's own reported quantity. On these two corpora the quantity the theory governs and the reported quantity move together.

The mismatch bound holds and is loose: at the best ridge of every lane, η measures between 0.01 and 1.61 against a bound that exceeds it by a factor between 60 and 3358. It certifies nothing useful at this operating point. The quantity that does the certifying is the margin in (30), computed directly, per corpus and per seed — and it certifies the consistent arm. The pipeline's own arm is a different estimator, and the bound's transfer condition $\gamma > \eta$, with $\gamma = \sqrt{\mathcal{R}_0} - \sqrt{\mathcal{R}_1}$ the consistent arm's root-risk margin, is what would carry the certificate across. It holds on 0 of the 248 (lane, ridge) pairs and on 0 of the 31 lanes, which is to say nowhere: wherever the consistent arm has a positive margin at all, γ is at most 0.15 and η exceeds it by a factor of at least 4.1. The out-of-fold arm behaves accordingly (Table 18). At the ridge where the consistent arm is best it raises the test risk on every EMIT lane, by 11 to 74 percent of $\mathcal{R}_0$ on Y_1, Y_2 and Y_3, and on 10 of the 11 OCO-2 lanes; its best gain over the eight ridges is at most 1.0 percent on Y_1–Y_3, 3.1 percent on one Y_4 seed and negative on the other 4, and between -0.4 and 0.3 percent on OCO-2. Its physical error at the consistent arm's best ridge is worse than the mean's on 30 of the 31 lanes, by up to 29 percent of the mean's own error, and the sign of that change agrees with the sign of its squared-risk change on all 31 lanes. So at this operating point — an 8,000-row kernel stage, four folds, the length scale chosen on the consistent labels — the correction the pipeline fits does nothing on these lanes, while the consistent arm on the same rows gains 15.3 to 25.5 percent on the components where its margin is positive. The identity locates the gain; the mismatch is what stands between it and the deployed estimator, and neither the bound nor the data close that gap here.

Remark 11.1. The identity assumes nothing; the leave-one-out gate built on top of it in Section 13 carries the hypotheses. That gate can be stated under uniform sup-norm deletion stability of the mean and of the correction, with bounded-difference constants, or under the weaker integrated L^2 stability by way of the Efron–Stein inequality [20]; the second is the form used here. The fifth problem of Section 12 needs the same same-sample machinery for a quite different reason.

12 The fifth problem: growing-rank spikes and what pruning may assume

The source is a manuscript on growing-rank spiked singular values and vectors with applications to pruning deep networks, and the theorem about it established in [45]. The model is $W_n = X_n + P_n$ with $P_n = U_n \Theta_n V_n^*$ of rank $r_n \to \infty$, $r_n = o(n)$, and $n/m \to c \in (0, 1]$, a growing-rank form of the spiked models of [4, 8]. Everything the source proves about cluster projectors and mixed dyads — and therefore everything it says about pruning — rests on a compression hypothesis for the resolvent compressed to the spike frame: writing

$$A_n(z) = U_n^* z (z^2 - X_n X_n^*)^{-1} U_n, \qquad D_n(z) = V_n^* z (z^2 - X_n^* X_n)^{-1} V_n,$$
$$B_n(z) = U_n^* X_n (z^2 - X_n^* X_n)^{-1} V_n,$$

the hypothesis asks, in operator norm and uniformly on a contour Γ off the spectrum, that $A_n \approx \phi_n I$, $D_n \approx \widetilde{\phi}_n I$ and $B_n \approx 0$, where ϕ_n and $\widetilde{\phi}_n$ are the corresponding normalized traces. The

source verifies the hypothesis in the Haar class. The theorem extends it to entries that are i.i.d., symmetric and obey a dimension-free Euclidean concentration inequality, throughout $r_n = o(n)$ (claim P19-IID-GC-BLOCK-02 in [45]), with

$$\beta_n(\Gamma) \;:=\; \sup_{z\in\Gamma}\Big[\|A_n-\phi_n I\|_{\mathrm{op}}+\|D_n-\widetilde{\phi}_n I\|_{\mathrm{op}}+\|B_n\|_{\mathrm{op}}\Big] \;=\; O_{\mathbb{P}}\!\left(\sqrt{\frac{r}{n}}+\sqrt{\frac{\log n}{n}}\right); \qquad (31)$$

up to constants and the contour net; for independent entries with bounded moments the anisotropic local law of Knowles and Yin [32] supplies the compression for deterministic frames at $r = o(\sqrt{n})$. An extension to centred laws without the symmetry is recorded in the same track's ledger without an archived proof. The concentration hypothesis remains, and nothing in the theorem applies to a pruning frame selected from the same matrix it is applied to.

12.1 What is measured

Three things. The entry law is the dial. A centred Bernoulli(0.2) variable is bounded, so it satisfies the concentration hypothesis by Talagrand's inequality [52], and it is as asymmetric as a two-point law with unit variance can be, with mass 0.8 at $-1/2$ and mass 0.2 at 2; a Student t_5 variable is exactly symmetric and has no exponential moment, so it fails the concentration hypothesis; a centred exponential variable fails both. Rademacher and Gaussian entries satisfy both. If the t_5 rows depart from the Rademacher rows, the concentration hypothesis is load-bearing at these sizes; if the Bernoulli rows depart, the symmetry is.

Everything is read off one thin singular value decomposition per design. With $X = \sum_k s_k w_k \tilde{v}_k^*$ and $\bar{U} = W^\top U$, $\bar{V} = \tilde{V}V$, the three blocks are $\bar{U}^*\mathrm{diag}\big(z/(z^2-s_k^2)\big)\bar{U}$, $\bar{V}^*\mathrm{diag}\big(z/(z^2-s_k^2)\big)\bar{V} + z^{-1}(I-\bar{V}^*\bar{V})$ and $\bar{U}^*\mathrm{diag}\big(s_k/(z^2-s_k^2)\big)\bar{V}$, so a whole contour costs nothing once the decomposition is in hand. The contour is a circle in the right half plane standing off the observed edge by fifteen percent of it, sampled at 64 points, and $\phi_n, \widetilde{\phi}_n$ are the empirical traces the source itself centres at. Table 20 reports β_n and, beside it, β_n divided by the rate in (31): if that rate is right the ratio does not run away with r or with n. The three blocks contribute to β_n in fixed proportion at aspect $1/2$, the A block 0.33 to 0.39, the D block 0.20 to 0.22 and the cross block B 0.39 to 0.46 of the sum, for every law, n and r run; the off-diagonal block, which the hypothesis wants to vanish, is the largest term at every size.

The third measurement is the adaptive spike. The theorem, like the source, assumes the spike frame is deterministic or independent of the noise. A rank-one perturbation built from the *observed* top singular pair of the same X leaves the empirical spectral distribution where it was, because a rank-one change cannot move it by more than $1/n$, while behaving nothing like an independent spike. The two arms are run side by side at matched θ — same matrix, same rank, same strength, and the only difference is whether the spike frame was drawn before looking at X or after.

12.2 What the entry law does, and what it does not

At $n = 1000$ the five entry laws agree to the precision of the measurement at every rank (Table 20). The asymmetric bounded law and the Rademacher law give $\beta_n = 0.0572 \pm 0.0020$ against 0.0573 ± 0.0011 at $r = 4$, 0.1354 ± 0.0016 against 0.1356 ± 0.0015 at $r = 16$ and 0.8104 ± 0.0047 against 0.8155 ± 0.0037 at $r = 256$, standard errors over 20 and 60 designs at $r = 16$. Symmetry enters the proof at one identifiable place: the expectation of the clipped self-resolvent is made an exact scalar matrix by invariance of the law under independent signed row and column permutations, whose commutant is the scalars; without the signs the commutant gains the all-ones matrix, and

Table 20: The compression block on a contour off the spectrum, at aspect $c = n/m = 0.5$ unless the row says otherwise. β_n is the supremum in (31) over 64 contour points; the ratio divides it by $\sqrt{r/n} + \sqrt{\log n/n}$. Entries are mean $\pm$ standard deviation over independent designs. Rows marked † lie outside the hypotheses of the theorem: Bernoulli(0.2) is bounded but not symmetric, t_5 is symmetric but not concentrated, and the exponential law is neither.

entry law	n	r	β_n	$\beta_n/(\sqrt{r/n} + \sqrt{\log n/n})$	designs
Gaussian	1000	4	0.06 ± 0.01	0.40 ± 0.05	40
Gaussian	1000	16	0.14 ± 0.01	0.65 ± 0.04	40
Gaussian	1000	64	0.32 ± 0.01	0.96 ± 0.04	40
Gaussian	1000	256	0.82 ± 0.02	1.39 ± 0.04	40
Rademacher	500	16	0.20 ± 0.02	0.69 ± 0.06	20
Rademacher	1000	4	0.06 ± 0.01	0.39 ± 0.05	40
Rademacher	1000	16	0.14 ± 0.01	0.65 ± 0.05	60
Rademacher	1000	64	0.32 ± 0.01	0.94 ± 0.03	60
Rademacher	1000	256	0.82 ± 0.02	1.38 ± 0.04	40
Rademacher	2000	16	0.09 ± 0.01	0.62 ± 0.04	20
Rademacher	2000	32	0.14 ± 0.00	0.77 ± 0.02	16
Rademacher	2000	128	0.33 ± 0.01	1.03 ± 0.03	16
Rademacher	4000	16	0.07 ± 0.00	0.60 ± 0.02	8
Rademacher	4000	64	0.14 ± 0.00	0.84 ± 0.02	8
Rademacher	4000	256	0.33 ± 0.00	1.10 ± 0.01	8
Bernoulli(0.2)†	500	16	0.20 ± 0.02	0.69 ± 0.05	20
Bernoulli(0.2)†	1000	4	0.06 ± 0.01	0.39 ± 0.06	20
Bernoulli(0.2)†	1000	16	0.14 ± 0.01	0.65 ± 0.03	20
Bernoulli(0.2)†	1000	64	0.32 ± 0.01	0.94 ± 0.04	20
Bernoulli(0.2)†	1000	256	0.81 ± 0.02	1.38 ± 0.04	20
Bernoulli(0.2)†	2000	16	0.10 ± 0.00	0.65 ± 0.03	20
Bernoulli(0.2)†	4000	16	0.07 ± 0.00	0.62 ± 0.02	8
Bernoulli(0.2)†, aspect 1	1000	16	0.20 ± 0.01	0.93 ± 0.06	20
exponential†	1000	4	0.06 ± 0.01	0.40 ± 0.05	20
exponential†	1000	16	0.13 ± 0.01	0.64 ± 0.04	20
exponential†	1000	64	0.32 ± 0.01	0.94 ± 0.03	20
exponential†	1000	256	0.81 ± 0.02	1.38 ± 0.03	20
Student t_5†	500	16	0.19 ± 0.03	0.64 ± 0.09	20
Student t_5†	1000	4	0.06 ± 0.01	0.41 ± 0.06	20
Student t_5†	1000	16	0.14 ± 0.01	0.65 ± 0.04	20
Student t_5†	1000	64	0.32 ± 0.01	0.94 ± 0.04	20
Student t_5†	1000	256	0.76 ± 0.09	1.29 ± 0.15	20
Student t_5†	2000	16	0.09 ± 0.01	0.61 ± 0.08	20
Student t_5†	4000	16	0.07 ± 0.00	0.61 ± 0.03	8
Student t_5†, aspect 1	1000	16	0.20 ± 0.01	0.93 ± 0.06	20

the expectation acquires a rank-one part whose contribution to the compressed block is of order r times its coefficient. What is measured is the hypothesis as the source states it, on the unclipped resolvent centred at the empirical trace, which absorbs any scalar part and leaves precisely the non-scalar bias visible; and that bias does not reach the size of the fluctuation at these r and n. The bias that a proof without the symmetry would have to control is not visible here. The heavy-tailed t_5 rows, outside the theorem, are indistinguishable from the Rademacher rows except at $r = 256$, where their spread across designs is four times larger, and the exponential rows, outside it on both counts, are indistinguishable throughout. At these sizes neither hypothesis of the theorem is visible in the compression error.

12.3 The rate

The theorem is an upper bound with an unspecified constant, and only its shape can be tested. At fixed $n = 1000$ the exponent of β_n in r reads 0.62 from $r = 4$ to 16, 0.61 from 16 to 64 and 0.68 from 64 to 256 for the Rademacher rows, and 0.62, 0.61, 0.68 for the Bernoulli rows, against the 1/2 the rate carries. Part of the excess at small r is the finite-r approach of an $r \times r$ random matrix's operator norm to its $2\sqrt{r}$ asymptote from below: measured directly on the three shapes that occur — real symmetric, complex symmetric, complex Ginibre — the operator norm divided by $2\sqrt{r}$ runs 0.77–0.85 at $r = 4$, 0.91–0.94 at $r = 16$, 0.97–0.98 at $r = 64$ and 0.99 at $r = 256$, so a quantity growing exactly like $\sqrt{r}$ would be measured with an apparent exponent between 0.59 and 0.62 from $r = 4$ to 16, near 0.53 from 16 to 64, and near 0.51 from 64 to 256. The first step is consistent with that; the second and third are not, and the third rises where the finite-r correction says it should fall. One confound has to be separated before that is read as a statement about the bound: at $r = 256$ and $n = 1000$ the ratio r/n is 0.256, outside the regime $r_n = o(n)$ the theorem is about, and there $\sqrt{r/n} = 0.51$ is itself of order one.

The experiment that separates the two is a ray of fixed r/n, along which the rate is constant up to the $\sqrt{\log n/n}$ term (Table 21). Along each ray β_n does not fall: it rises slightly and then flattens, while the rate itself falls by a fifth and an eighth over the same span because its $\sqrt{\log n/n}$ term is shrinking. So that term is not describing anything at these sizes: at fixed ratio the compression error converges to a constant rather than decaying with n. What it converges to is worth writing down. Subtracting the leading term $\sqrt{r/n}$ from the converged values at $n = 4000$ leaves an excess of 1.12 and 1.17 times r/n on the two rays, so

$$\beta_n(\Gamma) \;\longrightarrow\; \sqrt{\frac{r}{n}} \;+\; c\,\frac{r}{n}, \qquad c \approx 1.14, \tag{32}$$

with a leading coefficient consistent with exactly one. Fitted on those two points alone, (32) predicts the lanes it did not see: 0.1448 against 0.1441 measured at $(r, n) = (32, 2000)$, 0.3261 against 0.3254 at $(128, 2000)$, and 0.0678 against 0.0654 at $(16, 4000)$ — within a percent on the first two and 3.7 percent on the third, the smaller-n points sitting below the law because they have not finished rising.

In the regime the theorem is about, $r_n = o(n)$, the second term of (32) is $o(\sqrt{r/n})$ and the bound's rate is the truth — with, on this evidence, constant one rather than an unspecified C. The exponent above 1/2 measured at fixed n is the sum of the two terms of (32) read over a range of r where r/n climbed to 0.256 and the second term was no longer small. The sharper statement the measurements support is not that the compression error is $O(\sqrt{r/n} + \sqrt{\log n/n})$ but that it approaches $\sqrt{r/n} + c\,r/n$, with no visible $\log n$ at all.

Table 21: Rademacher rows along two rays of fixed r/n, at aspect $c = n/m = 0.5$. β_n is mean $\pm$ standard error over the designs; the excess is $\beta_n - \sqrt{r/n}$ and the last ratio divides it by r/n.

r	n	r/n	β_n	$\sqrt{r/n}$	excess	excess$/(r/n)$	designs
16	1000	0.016	0.1356 ± 0.0015	0.1265	+0.0092	0.57	60
32	2000	0.016	0.1441 ± 0.0011	0.1265	+0.0176	1.10	16
64	4000	0.016	0.1444 ± 0.0012	0.1265	+0.0179	1.12	8
64	1000	0.064	0.3176 ± 0.0013	0.2530	+0.0646	1.01	60
128	2000	0.064	0.3254 ± 0.0022	0.2530	+0.0724	1.13	16
256	4000	0.064	0.3276 ± 0.0014	0.2530	+0.0746	1.17	8

12.4 The frame you chose after looking

Table 22 is the counterexample with numbers on it. The two arms differ in one respect only: whether the rank-one spike's frame was drawn before or after seeing X. Below the transition the independent spike does what the independent-spike theory [8] says it does — at $\theta = 0.6$ and 0.8 the overlap with the planted direction is 0.003 and 0.006, which is to say nothing is recovered — while the frame read off X has overlap 1.000 to three decimals at every θ, with an outlier sitting at the unperturbed edge plus θ exactly. And the bulk cannot tell the two apart: the Kolmogorov–Smirnov distance between the perturbed and unperturbed spectra is $1.00/n$ in both arms at every θ, which is the smallest value the statistic can take on a sample of this size. Both arms are rank one, both leave a textbook Marchenko–Pastur bulk [38], and one of them recovers a direction that was never planted.

Table 22: The same rank-one spike, drawn independently of X and drawn from the observed top singular pair of X. Outlier is the leading singular value of $X + \theta uv^*$; overlap is $|\langle \hat{u}_1, u \rangle|^2$; the last two columns are the Kolmogorov–Smirnov distance between the perturbed and unperturbed bulk — everything except the leading singular value — in units of $1/n$, which is the resolution of the statistic on a sample of this size. Entries are mean $\pm$ standard deviation over independent designs; Rademacher entries throughout.

		independent frame		frame read off X		$n\times$ bulk KS		
n	θ	outlier	overlap	outlier	overlap	indep.	adapt.	designs
1000	0.6	2.41 ± 0.01	0.003 ± 0.004	3.01 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	0.8	2.41 ± 0.01	0.006 ± 0.009	3.21 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	1	2.41 ± 0.01	0.020 ± 0.022	3.41 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	1.2	2.42 ± 0.01	0.078 ± 0.050	3.61 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	1.4	2.44 ± 0.02	0.269 ± 0.085	3.81 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	1.6	2.51 ± 0.02	0.479 ± 0.032	4.01 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	2	2.73 ± 0.03	0.695 ± 0.014	4.41 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	2.5	3.08 ± 0.03	0.817 ± 0.007	4.91 ± 0.01	1.000 ± 0.000	1.00	1.00	20
1000	3	3.49 ± 0.03	0.877 ± 0.004	5.41 ± 0.01	1.000 ± 0.000	1.00	1.00	20
2000	1	2.41 ± 0.01	0.006 ± 0.011	3.41 ± 0.01	1.000 ± 0.000	1.00	1.00	12
2000	1.4	2.45 ± 0.01	0.338 ± 0.030	3.81 ± 0.01	1.000 ± 0.000	1.00	1.00	12
2000	2	2.75 ± 0.01	0.702 ± 0.013	4.41 ± 0.01	1.000 ± 0.000	1.00	1.00	12

The practical reading follows. A pruning frame taken from the singular vectors of the matrix being pruned is the second arm, not the first, and no amount of agreement between the observed singular-value histogram and its Marchenko–Pastur prediction licenses the independent calculation;

the bulk is exactly the statistic that cannot see the difference. What is needed is a frame built without looking, or a stability argument connecting the learned frame to one — the same same-sample problem that Section 11 handles for a mean and a correction. One more thing the frame is not: a substitute for the ridge. For kernel ridge regression truncated to the top r eigenvalues of the Gram matrix, Amini [1] gives the worst-case risk over the unit ball of the RKHS exactly, $\max\{H_r(\lambda), \mu_{r+1}\} + \frac{\sigma^2}{n}\sum_{i\le r}\left(\frac{\mu_i}{\mu_i+\lambda}\right)^2$ with $H_r(\lambda) = \max_{i\le r}\lambda^2\mu_i/(\mu_i+\lambda)^2$, and shows that truncating at any r at or above the level the ridge itself resolves never raises the minimax risk and strictly lowers it whenever $\mu_{r+1} > 0$; the implicit regularisation of the cut and the explicit one of λ are both needed. A pruning theory that assumes the ridge away, or a ridge path that assumes the frame away, is describing half of the estimator.

Remark 12.1. The object the fifth problem describes is easy to see in a trained network. For five weight matrices of a vision transformer trained on ImageNet (timm's `vit_base_patch16_224`, the first, seventh and last MLP input layers, 3072×768, one attention projection 2304×768 and one output projection 768×768), the number of singular values above the Marchenko–Pastur edge computed from each matrix's own entry standard deviation is 67, 78, 36, 78 and 2 of 768, against 0 of 768 for i.i.d. Gaussian matrices of the same shapes. This is one checkpoint, the edge is estimated from a matrix that contains the spikes and is therefore conservative, and the frame those singular vectors would supply is the second arm of Table 22 by construction: it shows the object, not the theorem.

13 Regularisation, overfitting, and a learned mean, on two corpora

Targets 2 and 3 ask when an interpolating kernel estimator is harmless. The same question can be put to data along the whole regularisation path rather than at its ridgeless end, with the kernel placed on top of a network rather than standing alone. It is put here to two corpora: the one-dimensional Burgers problem of PDEBench [51] and the advection problem of the operator-learning benchmark of de Hoop, Huang, Qian and Stuart [14], a setting in which kernel methods are competitive with neural operators [7] and Gaussian processes solve the underlying equations directly [12]. The evidence is thin and is reported as such: one seed of the regularisation path and one of the network-mean gate on Burgers, three seeds of the constant-mean gate there, two seeds of the dimension-gap diagnostic on Burgers and one on advection. None of it is a population statement; each table is one or two draws.

13.1 The path as a learning curve

For each corpus and sample size the ridge estimator is fitted at thirteen values of λ from 10^{-12} to 1 and its test risk recorded; the interpolant is the $\lambda \to 0$ end of the path. Two exponents are read off the same Gram matrix and labels the path was fitted on: the capacity $\hat{\alpha}$, the slope of log eigenvalue against log index, and the source $\hat{r}$, from the decay of the target's coefficients in the Gram eigenbasis — the source and capacity conditions of [11]. In the parametrisation of Cui, Loureiro, Krzakala and Zdeborová [13], ridge regression has a noiseless regime and a noisy one: the noiseless decay $n^{-2\alpha\min(r,1)}$ of the excess risk holds for n well below $n_1^* \asymp \sigma^{-1/(\alpha\min(r,1))}$, and the noisy optimal rate takes over for n well above $n_2^* \asymp \sigma^{-\max(2,\,1/(\alpha\min(r,1)))}$. The statistic that says whether interpolating was harmless is the ratio of the interpolant's risk to the best risk on the path.

On Burgers at $n = 4000$ with no label noise (Table 23) the interpolant is the optimum of the path — the ratio is 1.00 and the best ridge is the smallest one tried — with $\hat{\alpha} = 2.02$ and $\hat{r} = 0.63$,

Table 23: The regularisation path with no label noise, one seed. $\hat{\alpha}$ and $\hat{r}$ are the capacity and source exponents read off the Gram spectrum; the fourth column is $1/(\hat{\alpha}\min(\hat{r},1))$, the exponent of the noise level at which the noiseless decay ends in [13]; λ_* is the best ridge; the ratio is the interpolant's test risk over the best risk on the path, so 1.00 means interpolation was the optimum.

corpus	comp.	n	$\hat{\alpha}$	$\hat{r}$	exponent	λ_*	interp./best	seeds
burgers		4000	2.02	0.63	0.78	10^{-12}	1.00	1

Table 24: The ratio interpolant over best ridge as the label noise is turned up. Only the noiseless cell was run.

corpus	comp.	n	noise 0	0.03	0.1	0.3
burgers		4000	1.00	–	–	–

so that $1/(\hat{\alpha}\min(\hat{r},1)) = 0.78$: in the theory of [13] a tenfold increase in the noise moves the end of the noiseless decay by a factor of about 6 in n, and the takeover of the noisy rate, governed by $\max(2, 0.78) = 2$, by a factor of 100. Past the takeover the same theory fixes the shape of the ratio itself: with $\lambda \to 0$ the excess risk plateaus at the noise variance while the optimally regularised risk keeps falling like $n^{-2\alpha\min(r,1)/(1+2\alpha\min(r,1))}$, so the ratio should grow with n at that exponent, 0.72 for Burgers. The lanes at the other noise levels, which would test both predictions, were not run (Table 24 carries the one noiseless cell), so on this corpus the predictions stand as predictions.

13.2 The gate with a learned mean

The identity of Section 11 gates a kernel correction on top of a mean by leave-one-out risk. Here the mean is either a constant or a multilayer perceptron of width 256 trained for sixty epochs, so that the network's own fit and the correction's ridge λ enter together; leave-one-out, which stays consistent along iterative paths where generalised cross-validation [24] does not [44] and which is the selection principle of kernel flows [43, 27], is used throughout. The question is whether the gate keeps doing its two jobs once the mean has itself been fitted to the training labels: taking the gain when the correction helps, and refusing the step when it hurts. The correction the gate acts on is fitted to the residuals against the trained mean's own training-row predictions — the consistent arm of Section 11 — so the gains below, and the share of them the gate captures, are that arm's; the out-of-fold arm is not measured here.

On Burgers (Table 25) the network mean reaches a training error of 1.1×10^{-5} and the gate still takes 99 percent of the remaining gain with $\hat{\theta} = 0.99$; the mean's own risk of 1.5×10^{-5} falls to 3.5×10^{-6} against a best of 3.3×10^{-6}. On the advection suite with a constant mean the correction is the whole model, from a risk of 1.00 to 0.018 gated against a best of 0.015, and the gate takes it. On none of the five designs does the correction, gated or not, leave the test risk above the mean's own, so the gate's second job — refusing a harmful step — is not exercised on these corpora.

13.3 The order parameter on data that has none

The diagnostic of Section 8 maps the dimension gap onto a corpus through the Gram spectrum: d_{eff} is the number of directions resolved above the numerical floor, p_{eff} the participation ratio of the spectrum, standing in for the ambient dimension, and $\delta_{\text{eff}} = \log(d_{\text{eff}}/n)/\log p_{\text{eff}}$. Beside it stands the thing this is supposed to predict: the advantage of ridgeless interpolation over the best

Table 25: The leave-one-out gate on top of a constant mean and on top of a network mean. w is the network's weight decay; $\hat{\theta}$ the step the gate chooses; "captured" the share of the gain between the mean's risk and the best achievable risk that the gated step takes; "worse" the number of designs, gated and ungated, on which the correction left the test risk above the mean's own.

corpus	mean	w	$\hat{\theta}$	captured	worse (gated / full)	designs
burgers	const	–	0.95 ± 0.00	1.00 ± 0.00	0 / 0	3
burgers	mlp	0	0.99	0.99	0 / 0	1
caltech_adv	const	–	0.95	1.00	0 / 0	1

Table 26: The empirical dimension gap and the interpolation advantage $(R_{\text{best}} - R_{\text{interp}})/R_{\text{best}}$ on the two corpora, ordered by δ_{eff}; mean $\pm$ s.d. over the seeds where there are two.

corpus	comp.	n	d_{eff}	p_{eff}	δ_{eff}	advantage	seeds
caltech_adv		6000	5164	4.7	-0.096	-0.88	1
burgers		6000	5382	5.1	-0.067 ± 0.000	0.25 ± 0.01	2

ridge on a swept grid, as a fraction of the best risk, negative when interpolating loses (Table 26).

On the advection suite interpolating loses outright: at $n = 6000$ the best ridge, $\lambda = 10^{-5}$, has risk 0.0132 and the interpolant 0.0248, an advantage of -0.88, with $\delta_{\text{eff}} = -0.10$. On Burgers at the same n the interpolant wins by a quarter of the best risk on both seeds, at a δ_{eff} that is less negative. Two corpora are two points, and a rank correlation between δ_{eff} and the advantage across corpora — the statistic that would say whether the order parameter carries over to data — needs many more. The mapping from the sphere to a corpus is a modelling choice, and this is where it would be tested.

The theory side of Table 26 has been written down in the same coordinate. Lu, Lin, Xia and Huang [36] take $n \asymp d^{\gamma}$ on the sphere, a source parameter $s \geq 0$ for the smoothness of the target, and the ridge $\lambda \asymp d^{-u}$, and prove that overfitting is benign — in the under-regularised regime and at interpolation alike — exactly when $0 < s \leq \Gamma(\gamma)$, for a threshold function recalled in [36] from earlier work, where with $\ell = \lfloor \gamma \rfloor$

$$\Gamma(\gamma) = \frac{\gamma - \ell}{\ell} \quad (\ell < \gamma \leq \ell + \tfrac{1}{2}), \qquad \Gamma(\gamma) = \frac{\ell + 1 - \gamma}{\ell + 1} \quad (\ell + \tfrac{1}{2} < \gamma < \ell + 1),$$

$\Gamma = \infty$ for $\gamma \leq \frac{1}{2}$ and $\Gamma = 0$ at every integer γ. Since the harmonic block of degree k has dimension $N_{d,k} \asymp d^k$, the fractional part of γ is the dimension gap of Section 8 for the last resolved block, $\delta = k - \gamma$ at level k: the threshold closes when n sits at a block boundary, where Target 3's noise floor is exact, and it is widest when n sits at the geometric middle of two blocks. At the sizes of Section 7, $n = 2^d$ gives $\gamma = d \log 2 / \log d$, which is 3.35 at $d = 12$ and 3.51 at $d = 13$, so $\Gamma(\gamma)$ is 0.12 at both: in that asymptotic only targets of source below about 0.12 overfit benignly on those designs, with d far from the regime the theorem is stated in. In that reading δ_{eff} is an estimate of the coordinate on which the benignity threshold is a known function, and the taxonomy of benign, tempered and catastrophic overfitting [37] is a description of the same axis in the spectral coordinates of [13].

14 What the measurements say

1. The proved lower bound for Target 3 is conservative at reachable sizes by a factor between 1.7 and 2.1 at $h = 1/2$, between 5.5 and 6.1 at $h = 1$, and between 8.1 and 12.5 at $h = 2$, against the ten-window constant $c_{10} = 0.3106$, the strengthening of Theorem 4.1 quoted in Remark 4.5. Its constants are independent of bandwidth by construction while the measured functional is not; the gap between them is where a sharp, h-dependent constant would live. The single-block certificate the one-window argument bounds sits above its constant $c_1 = 5/36$ at every dimension and bandwidth measured, following the sawtooth of the block dimension rather than a trend, and the expectation bound on the captured leverage, (24), is the loose step at these sizes.

2. Within a harmonic block, at fixed critical degree, the noise risk falls as the dimension gap widens, at seven dimensions and one bandwidth and out to $\delta = 0.6$, with the few inversions confined to blocks of small harmonic dimension or to cells with two designs; the tail isotropy that Target 2's proof drives to zero follows the same ordering.

3. Both lemmas that Target 4's reduction names look true at finite size, one of them in a stronger form than the reduction requires, and the second with its constant identified rather than merely bounded: $d\ \mathrm{Tr}(A^4) \to$ a limit statistically indistinguishable from 2, which is the fourth moment of the semicircle law and therefore the value a uniformly random direction in Sym_d would give. The leave-one-out normal is generic at leading order; the Veronese structure of the rows appears only in the $1/d$ correction, where it is an order of magnitude larger than the generic one and is set by the rank-one dependence of the row, not by its kurtosis. After the reduction the matrix is consistent with the square Wishart hard edge at every dimension measured. The remaining difficulty is the rectangular hard edge for the traceless harmonic block itself, and one thing that is known not to be available on the way there is uniform integrability of the full inverse smallest singular value, which diverges at every fixed d.

4. The leverage concentration that lets Target 1's argument replace a correlated Stein comparison with second-moment matching shows $d\ \mathrm{Var}(q)$ of order one at $d = 64$ and $d = 128$, with a clear dependence on the augmentation strength, no dependence on the column normalization, and nothing that one doubling of d can say about the limit; the residual coefficients that argument needs have a one-line Hermite identification with nonnegativity built in (Proposition 10.1).

5. On the EMIT and OCO-2 corpora the same-sample correction identity is exact to 4×10^{-12}, and its sign is computable per component and per ridge. On two of the four EMIT components the margin is positive at every ridge on every seed; on the third it changes sign at small ridge on some seeds; on the fourth, where the mean is poor, it is negative almost everywhere and the correction does nothing; on the three OCO-2 bands the correction is close to neutral. Where the consistent arm helps, its gain in squared risk transfers to the pipeline's physical error one for one. The mismatch bound holds and is loose by two to three orders of magnitude, its transfer condition fails on every (lane, ridge) pair, and the arm the pipeline fits, to out-of-fold residuals, gains nothing on these lanes.

6. For the fifth problem, the compression error of the spike-frame resolvent is indistinguishable across five entry laws inside and outside the hypotheses of the theorem, at every rank and size run. Along rays of fixed r/n it approaches $\sqrt{r/n} + 1.14\, r/n$, with no visible $\log n$ term and

a leading coefficient of one; the exponent above $1/2$ seen at fixed n is the second term. Two rank-one spikes with bulk spectra that differ by the resolution of the statistic and nothing more have overlaps of 0.003 and 1.000 with the planted direction, according to whether the frame was drawn before or after looking at the matrix.

7. On the two corpora of the regularisation block, one seed at a time, the noiseless Burgers interpolant is the optimum of the path at $n = 4000$, and on the advection suite interpolating loses to the best ridge by 88 percent of the best risk, at a more negative empirical dimension gap; the leave-one-out gate keeps its first property when the mean it corrects is a trained network, taking 99 percent of the gain on Burgers, and its second property is not exercised.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

This paper is a product of an ongoing project on the use of AI systems for research mathematics, provisionally called Math Brain, which studies how to make such systems better at solving open problems; the project and its tooling will be released publicly in due course. The proofs of Part I were found by the automated programme ProofNet24 [45], whose workers run on OpenAI's Codex, and were rewritten, checked and typeset by Claude (Anthropic; Claude Fable 5.1). The experiments were designed, the code written and run, the records audited and this manuscript drafted by Claude (Claude Opus 5 and Claude Fable 5.1) and Codex (OpenAI) sessions between 14 and 16 September 2026. The author set the goals of the project, commissioned and directed the work and approved the release; the author did not carry out the mathematics, the computations or the writing, and takes responsibility for the decision to publish.

Data and code availability

The drivers, the 223 run records, the scripts of the laptop-scale computations quoted in Sections 9.3 and 12, and the build script that regenerates every table and quoted number from the run records and checks the identities (26), (22), (27), (30) and (28) against them are in the repository `yspennstate/math-open-problems`, under `paper9_kernel_open_problems`, together with the ProofNet24 run reports under `proofnet24`, one directory per track with every run preserved under its own timestamp. The repository is private and is available from the author on request.

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